

# Принципы воспроизводимого анализа данных

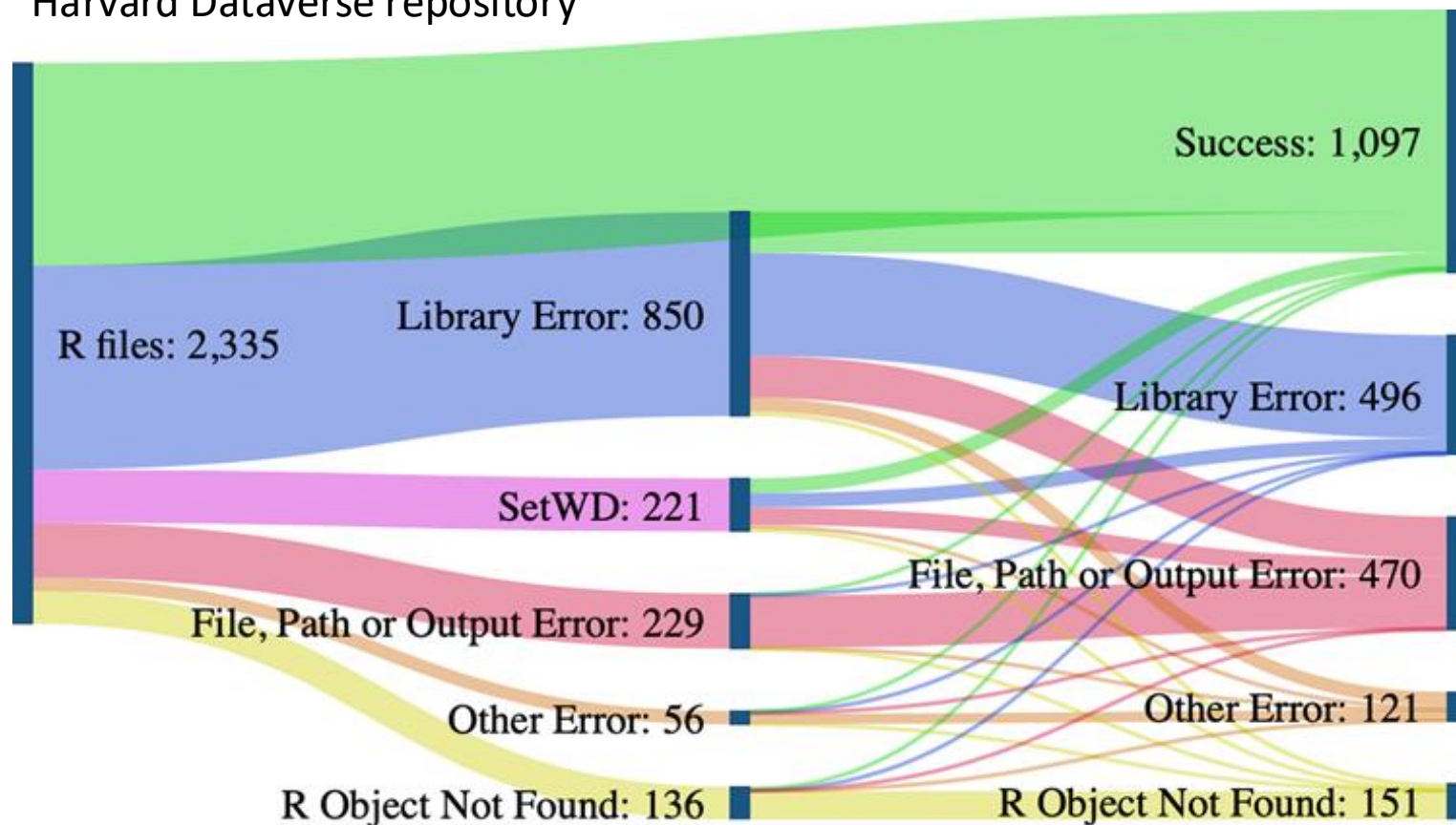
Kirill Voronin <imvoroninkir@gmail.com>

# План

- Основные проблемы и best practices
- Установка пакетов через renv
- Проверки (assert)
- Контроль версий при помощи Git
  
- Создадим проект используя Git и renv
  - Повторяемый Rmarkdown
  - Топ фильмов IMDB
- Запушим на GitHub
- Поработаем вместе используя GitHub pull request (PR)

# Воспроизводимость

Harvard Dataverse repository



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на самом деле ниже

Trisovic, A., Lau, M.K., Pasquier, T. *et al.* A large-scale study on research code quality and execution. *Sci Data* 9, 60 (2022). <https://doi.org/10.1038/s41597-022-01143-6>

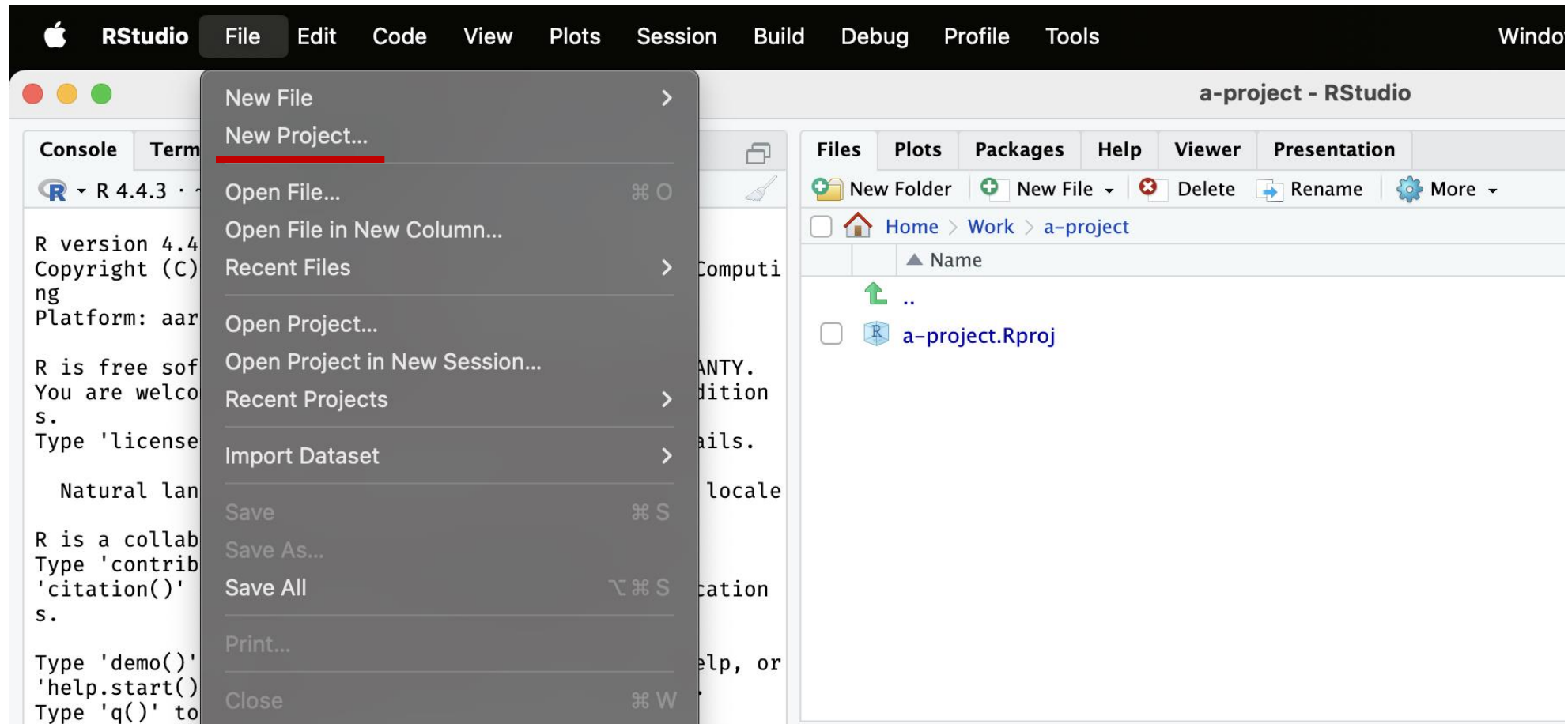
# Относительные пути

```
read.csv("C:\\Users\\Nastya\\Desktop\\my-project\\data\\patients.csv")
```

```
setwd("C:\\Users\\Nastya\\Desktop\\my-project")
```

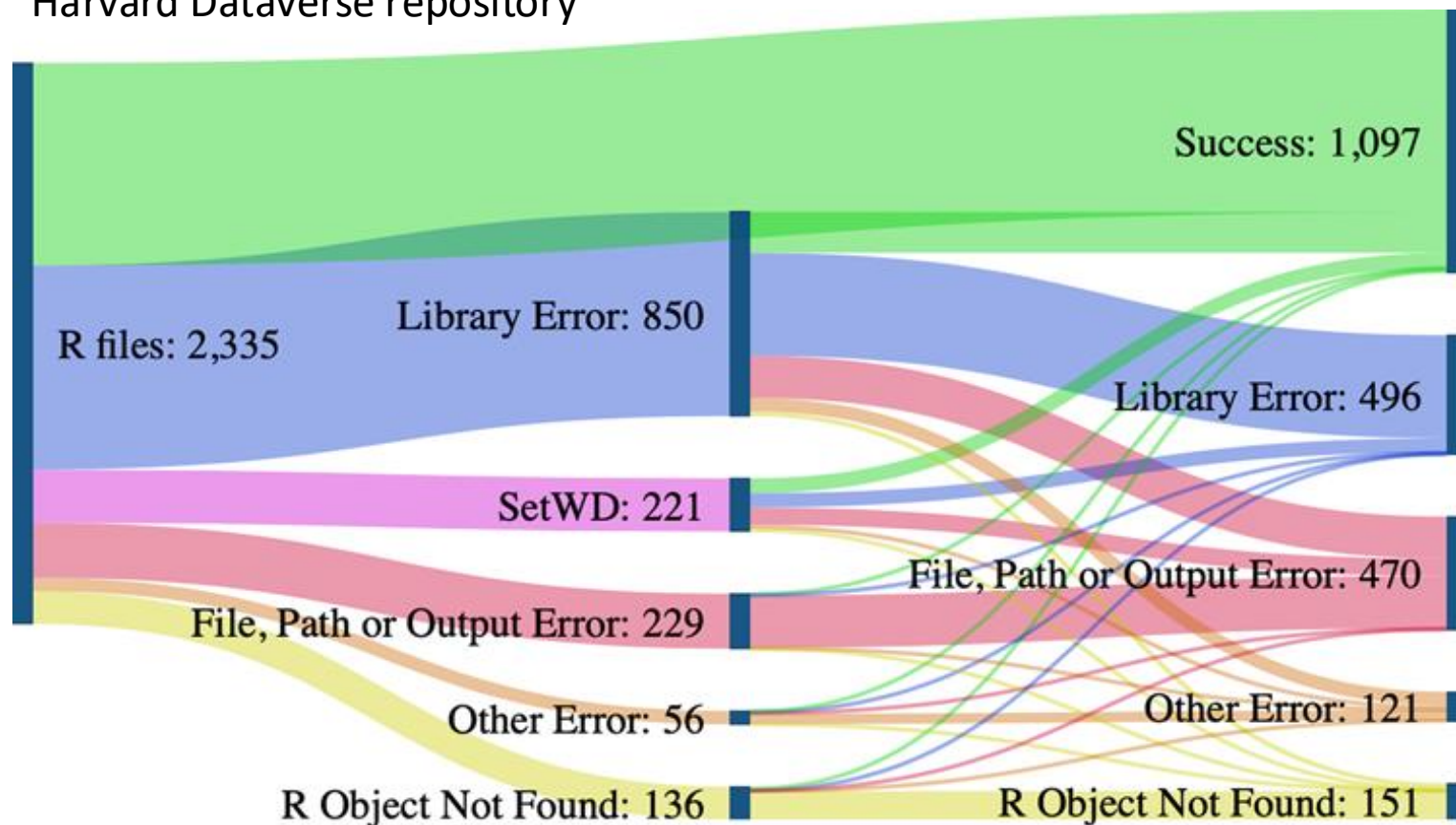
```
read.csv("data/patients.csv")
```

# Проекты в RStudio



# Воспроизводимость

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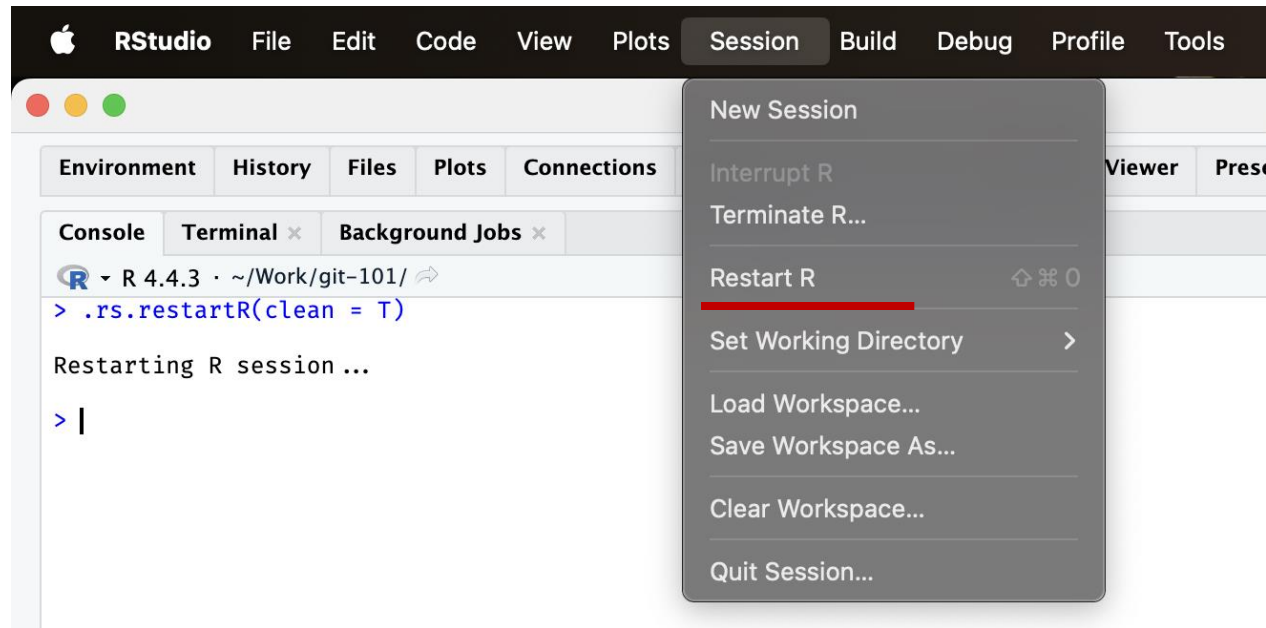


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на самом деле ниже

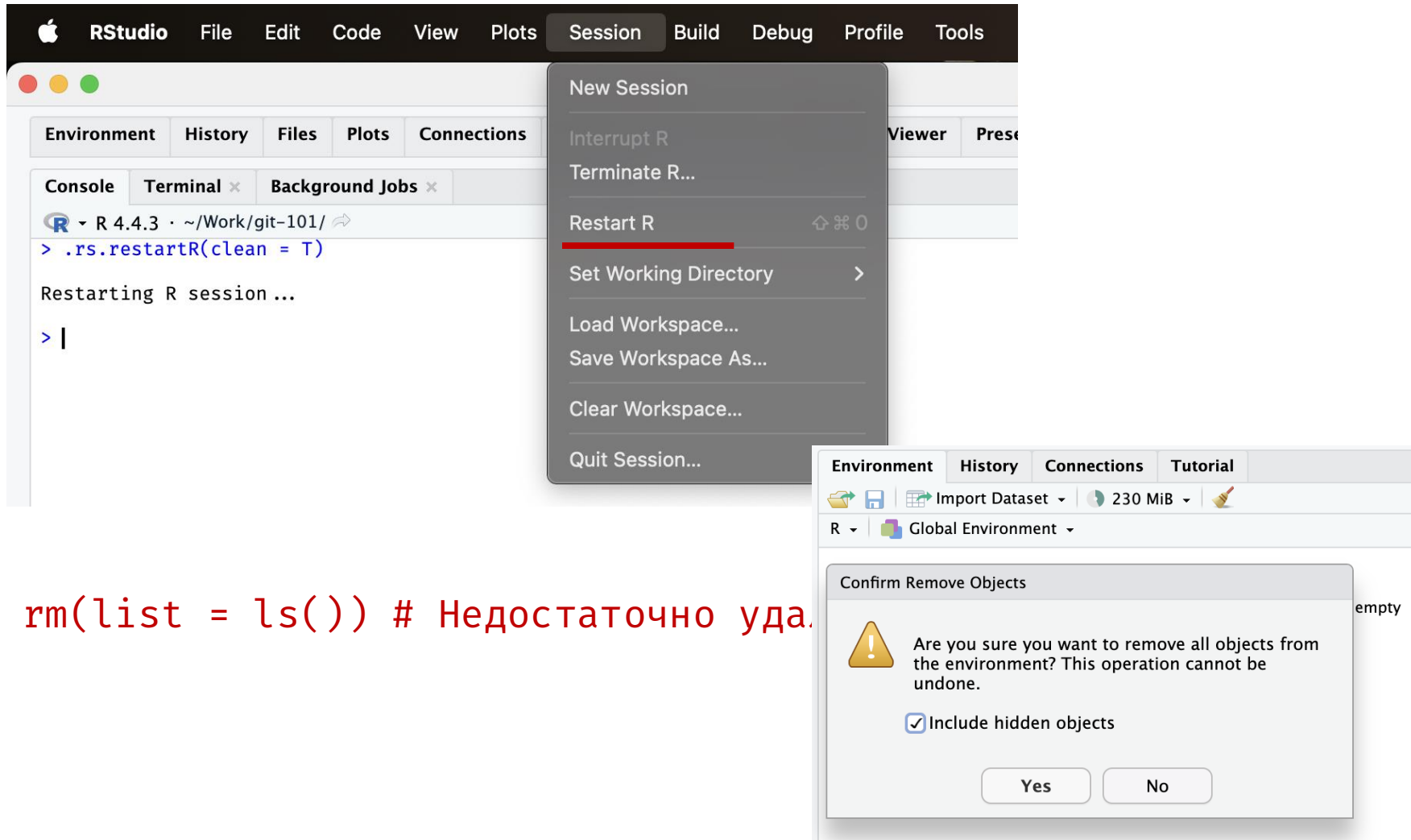
Trisovic, A., Lau, M.K., Pasquier, T. *et al.* A large-scale study on research code quality and execution. *Sci Data* 9, 60 (2022). <https://doi.org/10.1038/s41597-022-01143-6>

# Перезапускайте сессию



`rm(list = ls())` # Недостаточно удалить все переменные

# Перезапускайте сессию



The screenshot shows the RStudio application window. The top menu bar includes Apple logo, RStudio, File, Edit, Code, View, Plots, Session, Build, Debug, Profile, and Tools. The 'Session' menu is open, displaying options: New Session, Interrupt R, Terminate R..., Restart R (highlighted with a red underline), Set Working Directory, Load Workspace..., Save Workspace As..., Clear Workspace..., and Quit Session... The console on the left shows the command `> .rs.restartR(clean = T)` and the output 'Restarting R session ...'. Below the console, the text `rm(list = ls()) # Недостаточно уда.` is written in red. In the bottom right, a 'Confirm Remove Objects' dialog box is open, asking 'Are you sure you want to remove all objects from the environment? This operation cannot be undone.' with a checked option 'Include hidden objects' and 'Yes'/'No' buttons.

Environment History Files Plots Connections

Console Terminal x Background Jobs x

R 4.4.3 · ~/Work/git-101/

```
> .rs.restartR(clean = T)
```

Restarting R session ...

```
> |
```

Session Build Debug Profile Tools

New Session

Interrupt R

Terminate R...

Restart R

Set Working Directory

Load Workspace...

Save Workspace As...

Clear Workspace...

Quit Session...

Environment History Connections Tutorial

Import Dataset 230 MiB

R Global Environment

Confirm Remove Objects

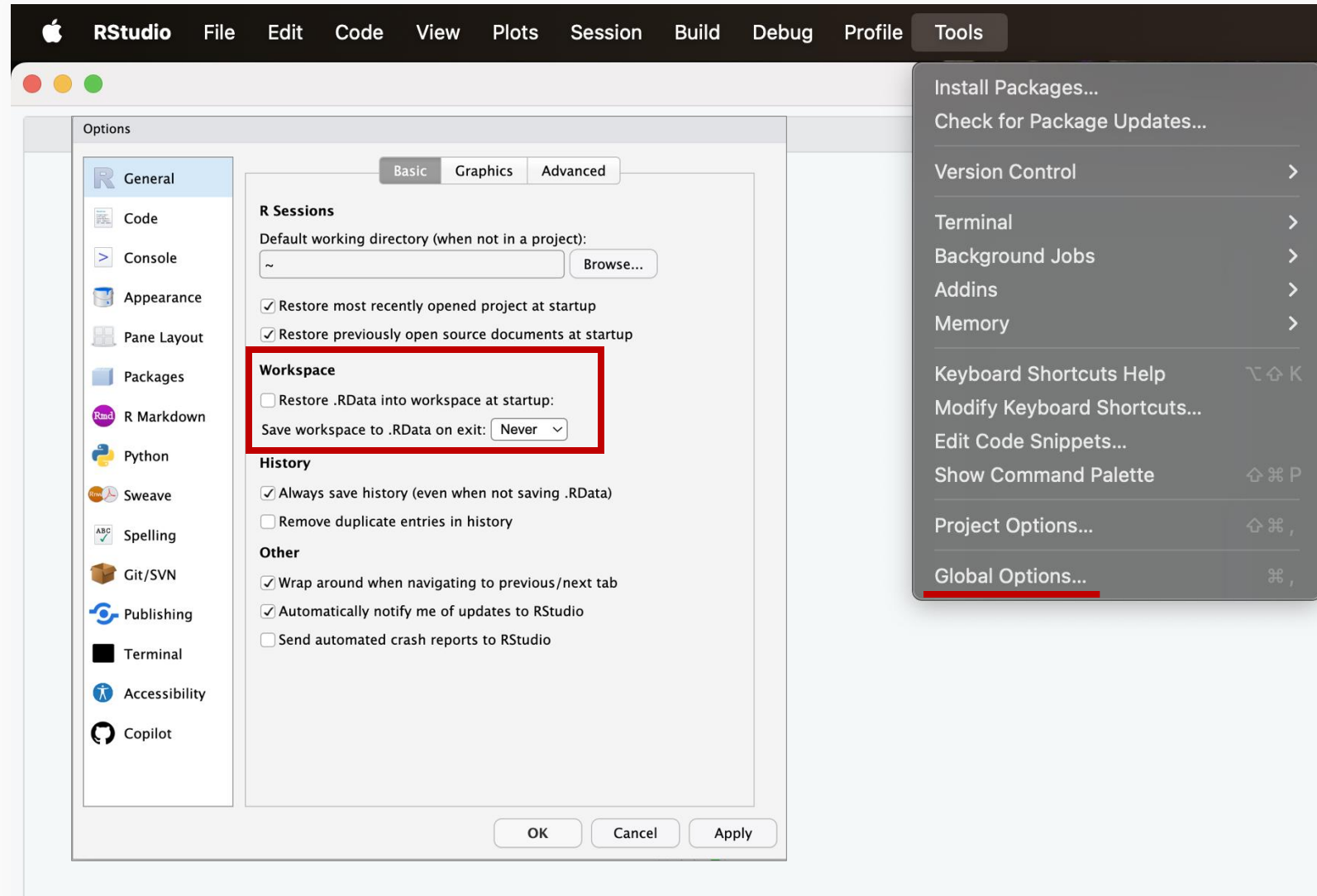
Are you sure you want to remove all objects from the environment? This operation cannot be undone.

☒ Include hidden objects

Yes No

`rm(list = ls()) # Недостаточно уда.`

# Чистая сессия



В директории проекта  
не должно быть файла

.RData  
(файл с состоянием)

# Сохранение объектов

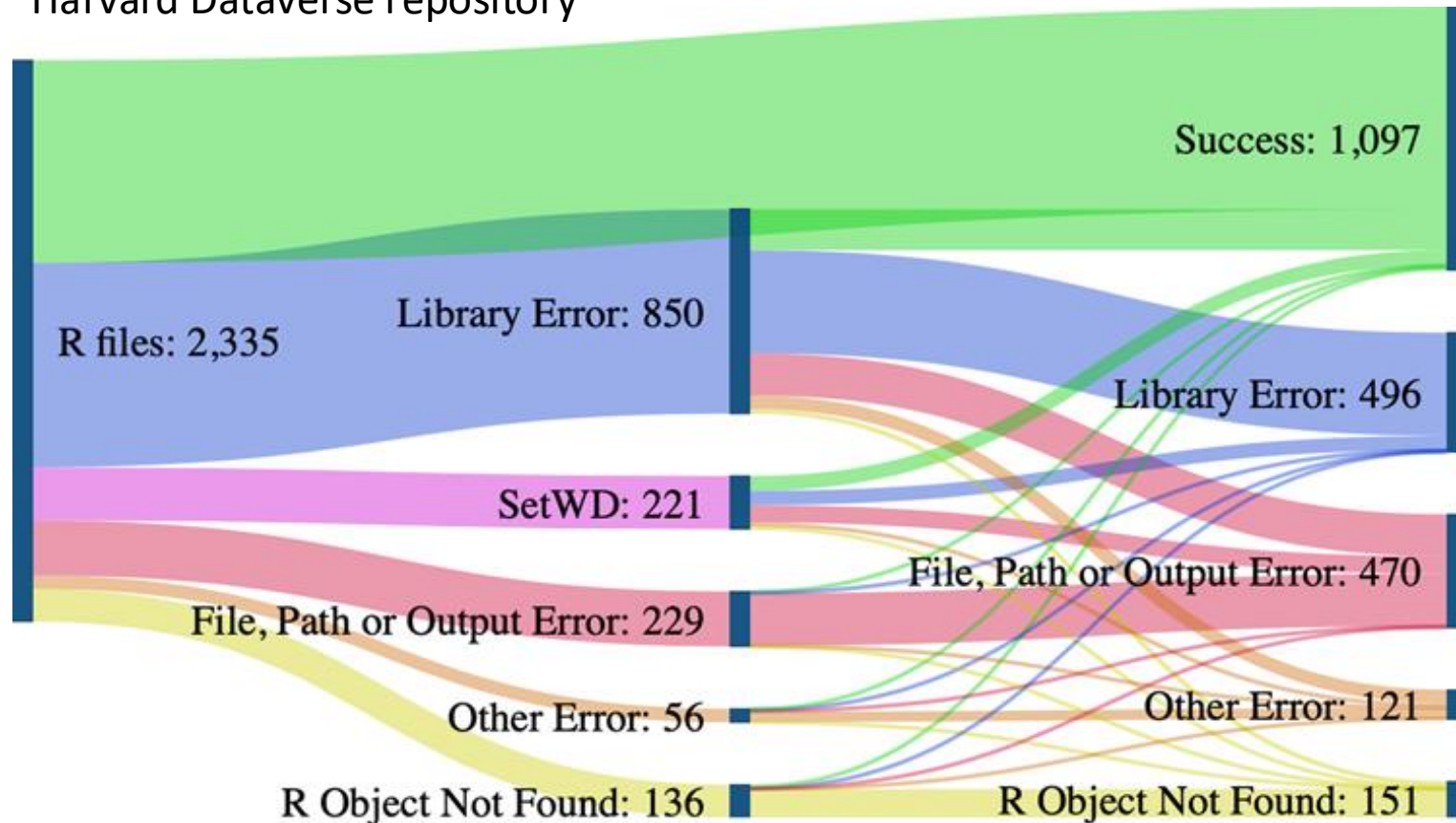
```
save(fit, file = "complex-fit.RData")  
load("complex-fit.Rdata")
```

```
saveRDS(fit, "complex-fit.rds")           # один объект за раз  
fit <- readRDS("complex-fit.rds")         # просто вернет значение
```

```
write_rds(fit, "complex-fit.rds")         # то же самое ↑  
fit <- read_rds("complex-fit.rds")
```

# Воспроизводимость

Harvard Dataverse repository



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на самом деле ниже

Trisovic, A., Lau, M.K., Pasquier, T. *et al.* A large-scale study on research code quality and execution. *Sci Data* 9, 60 (2022). <https://doi.org/10.1038/s41597-022-01143-6>

# install.packages

```
> install.packages(c("readr", "dplyr"))
```

```
Installing packages into '/Users/aavoronin/Library/R/x86_64/4.4/library'
```

```
(as 'lib' is unspecified)
```

```
also installing the dependencies 'bit', 'prettyunits', 'pkgconfig', 'fansi', 'bit64', 'withr', 'progress', 'utf8', 'cli', 'clipr', 'crayon', 'hms', 'lifecycle', 'R6', 'rlang', 'tibble', 'vroom', 'cpp11', 'tzdb', 'generics', 'glue', 'magrittr', 'pillar', 'tidyselect', 'vctrs'
```

# install.packages

```
> install.packages(c("readr", "dplyr"))
```

```
Installing packages into '/Users/aavoronin/Library/R/x86_64/4.4/library'
```

```
(as 'lib' is unspecified)
```

```
also installing the dependencies 'bit', 'prettyunits', 'pkgconfig', 'fansi', 'bit64', 'withr', 'progress', 'utf8', 'cli', 'clipr', 'crayon', 'hms', 'lifecycle', 'R6', 'rlang', 'tibble', 'vroom', 'cpp11', 'tzdb', 'generics', 'glue', 'magrittr', 'pillar', 'tidyselect', 'vctrs'
```

## 2.1.5

# install.packages

```
> install.packages(c("readr", "dplyr"))
```

```
Installing packages into '/Users/aavoronin/Library/R/x86_64/4.4/library'
```

```
(as 'lib' is unspecified)
```

```
also installing the dependencies 'bit', 'prettyunits', 'pkgconfig', 'fansi', 'bit64', 'withr', 'progress', 'utf8', 'cli', 'clipr', 'crayon', 'hms', 'lifecycle', 'R6', 'rlang', 'tibble', 'vroom', 'cpp11', 'tzdb', 'generics', 'glue', 'magrittr', 'pillar', 'tidyselect', 'vctrs'
```

2.1.5 → 3.0.4

# sessionInfo

```
sessionInfo()
```

```
R version 4.4.2 (2024-10-31)  
Platform: x86_64-apple-darwin20  
Running under: macOS Monterey 12.5
```

```
Matrix products: default
```

```
BLAS:   /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/lib/libRblas.0.dylib
```

```
LAPACK: /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.0
```

```
locale:
```

```
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
```

```
time zone: Europe/Moscow
```

```
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices datasets  utils      methods    base
```

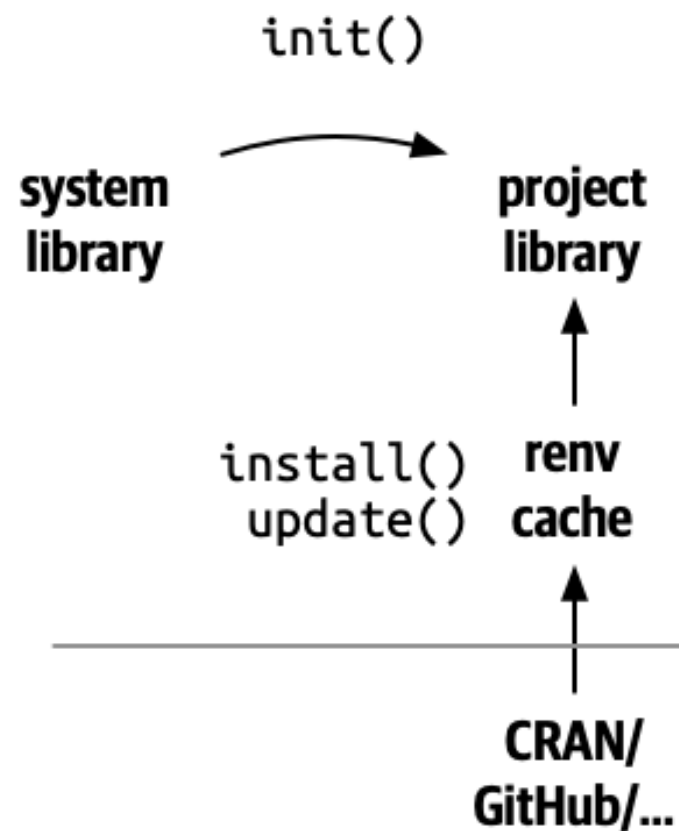
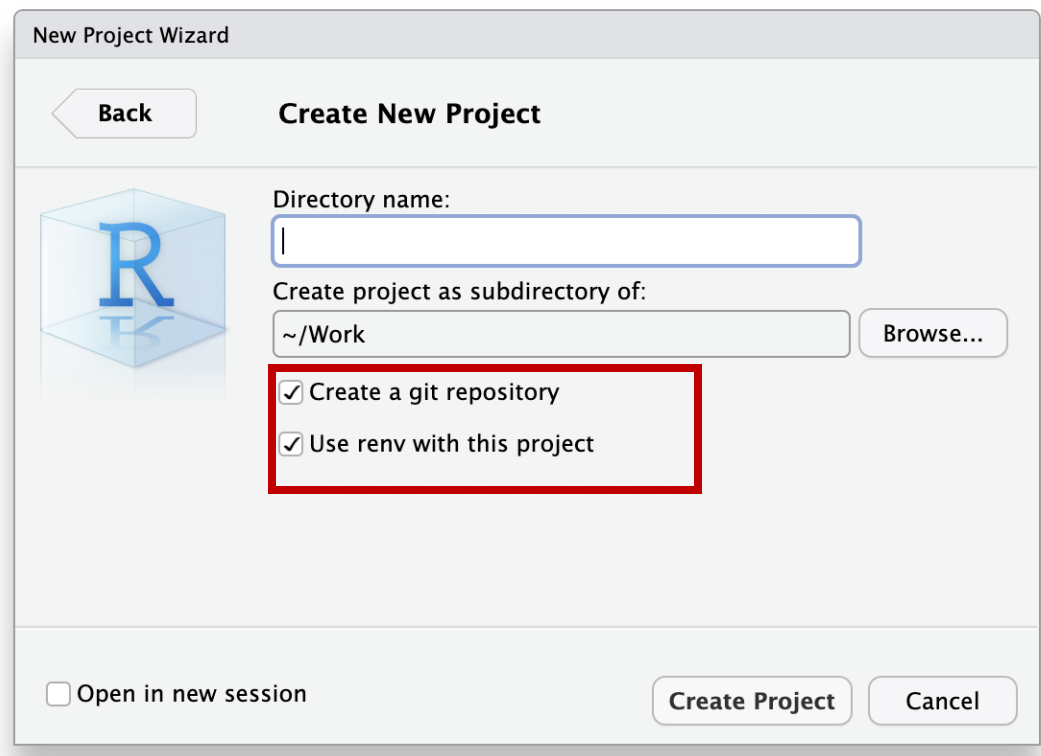
```
other attached packages:
```

```
[1] readr_2.1.5 dplyr_1.1.4
```

```
loaded via a namespace (and not attached):
```

|                      |                 |                 |                   |
|----------------------|-----------------|-----------------|-------------------|
| [1] crayon_1.5.3     | vctrs_0.6.5     | cli_3.6.3       | knitr_1.49        |
| [5] rlang_1.1.4      | xfun_0.49       | renv_1.0.11     | generics_0.1.3    |
| [9] jsonlite_1.8.9   | bit_4.5.0.1     | glue_1.8.0      | htmltools_0.5.8.1 |
| [13] sass_0.4.9      | hms_1.1.3       | fansi_1.0.6     | rmarkdown_2.29    |
| [17] evaluate_1.0.1  | jquerylib_0.1.4 | tibble_3.2.1    | tzdb_0.4.0        |
| [21] fastmap_1.2.0   | yaml_2.3.10     | lifecycle_1.0.4 | compiler_4.4.2    |
| [25] pkgconfig_2.0.3 | digest_0.6.37   | R6_2.5.1        | tidyselect_1.2.1  |
| [29] utf8_1.2.4      | parallel_4.4.2  | vroom_1.6.5     | pillar_1.9.0      |
| [33] magrittr_2.0.3  | bslib_0.8.0     | withr_3.0.2     | bit64_4.5.2       |
| [37] tools_4.4.2     | cachem_1.1.0    |                 |                   |

# renv



# renv

```
> renv::install(c("readr", "dplyr"))
```

The following package(s) will be installed:

- bit [4.5.0.1]
- bit64 [4.5.2]
- cli [3.6.3]
- clipr [0.8.0]
- cpp11 [0.5.1]
- crayon [1.5.3]
- dplyr [1.1.4]
- fansi [1.0.6]
- generics [0.1.3]
- glue [1.8.0]
- hms [1.1.3]
- lifecycle [1.0.4]
- magrittr [2.0.3]
- pillar [1.9.0]
- pkgconfig [2.0.3]
- prettyunits [1.2.0]
- progress [1.2.3]
- R6 [2.5.1]
- readr [2.1.5]
- rlang [1.1.4]
- tibble [3.2.1]
- tidyselect [1.2.1]
- tzdb [0.4.0]
- utf8 [1.2.4]
- vctrs [0.6.5]
- vroom [1.6.5]
- withr [3.0.2]

These packages will be installed into "~/Work/demo/renv/library/macos/R-4.4/x86\_64-apple-darwin20".

# renv.lock

```
> renv::install(c("readr", "dplyr"))
```

The following package(s) will be installed:

```
- bit          [4.5.0.1]
- bit64        [4.5.2]
- cli          [3.6.3]
- clipr        [0.8.0]
- cpp11        [0.5.1]
- crayon       [1.5.3]
- dplyr        [1.1.4]
- fansi        [1.0.6]
- generics     [0.1.3]
- glue         [1.8.0]
- hms          [1.1.3]
- lifecycle    [1.0.4]
- magrittr     [2.0.3]
- pillar       [1.9.0]
- pkgconfig    [2.0.3]
- prettyunits  [1.2.0]
- progress     [1.2.3]
- R6           [2.5.1]
- readr        [2.1.5]
- rlang        [1.1.4]
- tibble       [3.2.1]
- tidyselect   [1.2.1]
- tzdb         [0.4.0]
- utf8         [1.2.4]
- vctrs        [0.6.5]
- vroom        [1.6.5]
- withr        [3.0.2]
```

These packages will be installed into "~/Work/demo/renv/library/macos/R-4.4/x86\_64-apple-darwin20".

renv::snapshot()



```
"prettyunits": {"source": "CRAN", "version": "1.2.0"},
"progress": {"source": "CRAN", "version": "1.2.3"},
"readr": {
  "Package": "readr",
  "Version": "2.1.5",
  "Source": "Repository",
  "Repository": "CRAN",
  "Requirements": [
    "R",
    "R6",
    "cli",
    "clipr",
    "cpp11",
    "crayon",
    "hms",
    "lifecycle",
    "methods",
    "rlang",
    "tibble",
    "tzdb",
    "utils",
    "vroom"
  ],
  "Hash": "9de96463d2117f6ac49980577939dfb3"
},
"renv": {
  "Package": "renv",
  "Version": "1.0.11",
  "Source": "Repository",
  "Repository": "CRAN",
  "Requirements": [
    "utils"
  ],
  "Hash": "47623f66b4e80b3b0587bc5d7b309888"
},
"rlang": {"source": "CRAN", "version": "1.1.4"},
"tibble": {"source": "CRAN", "version": "3.2.1"},
"tidyselect": {"source": "CRAN", "version": "1.2.1"}
```

# renv.lock

```
> renv::install(c("readr", "dplyr"))
```

The following package(s) will be installed:

```
- bit          [4.5.0.1]
- bit64        [4.5.2]
- cli          [3.6.3]
- clipr        [0.8.0]
- cpp11        [0.5.1]
- crayon       [1.5.3]
- dplyr        [1.1.4]
- fansi        [1.0.6]
- generics     [0.1.3]
- glue         [1.8.0]
- hms          [1.1.3]
- lifecycle    [1.0.4]
- magrittr     [2.0.3]
- pillar       [1.9.0]
- pkgconfig    [2.0.3]
- prettyunits  [1.2.0]
- progress     [1.2.3]
- R6           [2.5.1]
- readr        [2.1.5]
- rlang        [1.1.4]
- tibble       [3.2.1]
- tidyselect   [1.2.1]
- tzdb         [0.4.0]
- utf8         [1.2.4]
- vctrs        [0.6.5]
- vroom        [1.6.5]
- withr        [3.0.2]
```

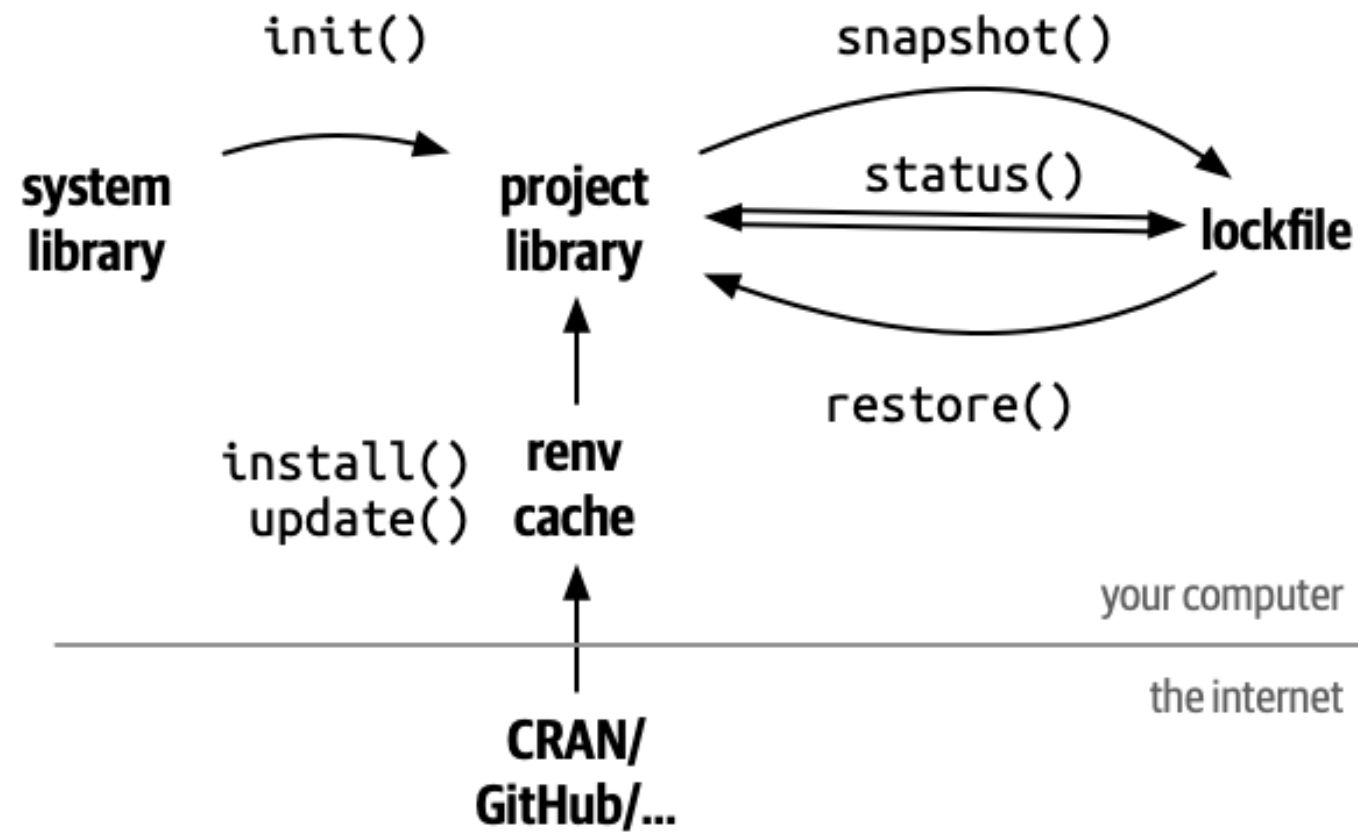
These packages will be installed into "~/Work/demo/renv/library/macos/R-4.4/x86\_64-apple-darwin20".

```
# или lock = TRUE
```

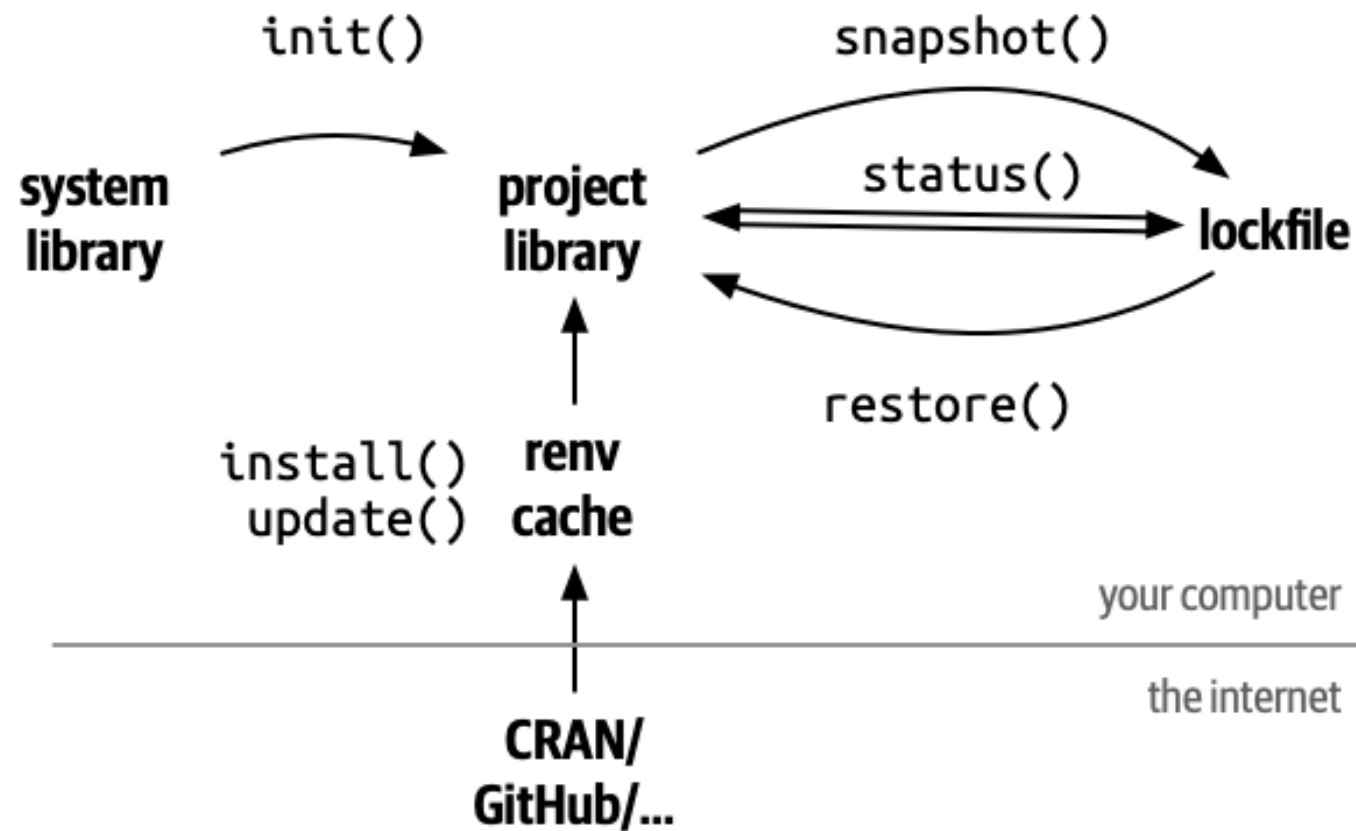
```
renv::install(
  c("readr", "dplyr"),
  lock = TRUE
)
```

```
"prettyunits": {,
"progress": {,
"readr": {
  "Package": "readr",
  "Version": "2.1.5",
  "Source": "Repository",
  "Repository": "CRAN",
  "Requirements": [
    "R",
    "R6",
    "cli",
    "clipr",
    "cpp11",
    "crayon",
    "hms",
    "lifecycle",
    "methods",
    "rlang",
    "tibble",
    "tzdb",
    "utils",
    "vroom"
  ],
  "Hash": "9de96463d2117f6ac49980577939dfb3"
},
"renv": {
  "Package": "renv",
  "Version": "1.0.11",
  "Source": "Repository",
  "Repository": "CRAN",
  "Requirements": [
    "utils"
  ],
  "Hash": "47623f66b4e80b3b0587bc5d7b309888"
},
"rlang": {,
"tibble": {,
"tidyselect": {,
```

# renv



# renv



rig – несколько версий R

# stop

```
set.seed(42)
df <- data.frame(x = sample(c("Lo", "Hi", NA), 10, replace = T))

if (any(is.na(df$x))) {
  stop("Risk group missing")
}

#> Error: Risk group missing
```

# stopifnot

```
set.seed(42)
df <- data.frame(x = sample(c("Lo", "Hi", NA), 10, replace = T))

stopifnot(is.na(df$x))
#> Error: is.na(df$x) are not all TRUE

stopifnot("Risk group missing" = is.na(df$x))
#> Error: Risk group missing
```

# assertr

```
library(assertr)
```

```
library(dplyr)
```

```
set.seed(42)
```

```
df <- data.frame(x = sample(c("Lo", "Hi", NA), 10, replace = T),  
                 y = rnorm(10))
```

```
df |>
```

```
  verify(!is.na(x)) |>
```

```
  count(x)
```

```
#> verification [!is.na(x)] failed! (2 failures)
```

```
#>
```

```
#>      verb redux_fn predicate column index value
```

```
#> 1 verify      NA !is.na(x)      NA      9     NA
```

```
#> 2 verify      NA !is.na(x)      NA     10     NA
```

```
#> Error: assertr stopped execution
```

# вставляется между пайпами

# любое логическое выражение

# assertr

```
df |>
```

```
  assert(not_na, x, y) |>  
  count(x)
```

# assert + предикат вместо verify

```
#> Column 'x' violates assertion 'not_na' 2 times
```

```
#>      verb redux_fn predicate column index value
```

```
#> 1 assert      NA    not_na      x      9  <NA>
```

```
#> 2 assert      NA    not_na      x     10  <NA>
```

```
#> Error: assertr stopped execution
```

# assertr

```
df |>
```

```
  assert(not_na, x, y, error_fun = just_warn) |>      # обработчик ошибок  
  count(x)
```

```
#> Column 'x' violates assertion 'not_na' 2 times
```

```
#>      verb redux_fn predicate column index value
```

```
#> 1 assert      NA    not_na      x      9  <NA>
```

```
#> 2 assert      NA    not_na      x     10  <NA>
```

```
#> Warning: assertr encountered errors
```

```
#>      x n
```

```
#> 1  Hi 3
```

```
#> 2  Lo 5
```

```
#> 3 <NA> 2
```

# assertr

```
df |>
```

```
  insist(within_n_sds(1), y)
```

```
#> Column 'y' violates assertion 'within_n_sds(1)' 4 times
```

```
#>      verb redux_fn      predicate column index      value
#> 1 insist      NA within_n_sds(1)      y      2  1.554896
#> 2 insist      NA within_n_sds(1)      y      3 -1.187641
#> 3 insist      NA within_n_sds(1)      y      5 -1.086133
#> 4 insist      NA within_n_sds(1)      y      6  1.613373
```

```
#> Error: assertr stopped execution
```

# Встроенные предикаты

`within_bounds(0,1)`      `# в диапазоне от 0 до 1`

`in_set("M","F")`      `# из множества M, F`

`within_n_sds(5)`      `# ± n SD`

`within_n_mads(5)`      `# ± n MAD`

`# Функция возвращая FALSE если условие не выполняется`

```
is_div <- function(d) {  
  function(x) x %% d == 0  
}
```

# assertr

```
library(survival)
```

```
fit_data <- aml |>  
  verify(time >= 0) |>  
  assert(in_set(0, 1), status)
```

```
fit <- survfit(Surv(time, status) ~ 1, fit_data)
```

# reprex (Reproducible Example)

```
set.seed(42)                                # Фиксируем ГСЧ
x <- rnorm(100)                              # Создаем нужную величину

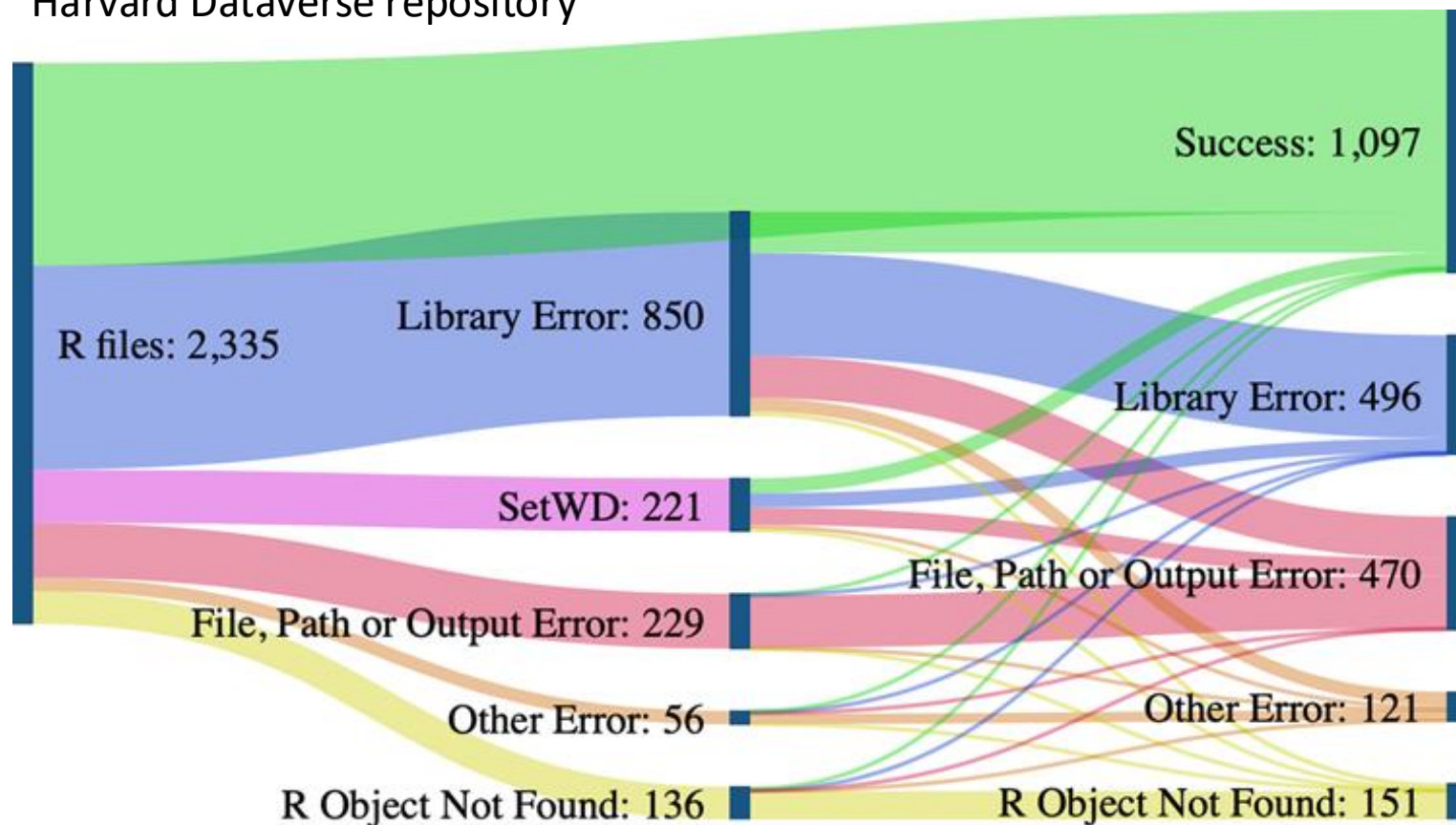
df <- tribble(~ x, ~ y,                     # или датасет
              "M", 12.3,
              "F", 45.6)

lm(mpg ~ wt + vs, mtcars)                   # сам код
# Coefficients:                             # в идеале с результатом
# (Intercept)          wt          vs      # в комментариях
#      33.004        -4.443         3.154      #

sessionInfo()                              # версии библиотек
# ...
# длинный код -> https://gist.github.com
```

# Воспроизводимость

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на самом деле ниже

Trisovic, A., Lau, M.K., Pasquier, T. *et al.* A large-scale study on research code quality and execution. *Sci Data* 9, 60 (2022). <https://doi.org/10.1038/s41597-022-01143-6>



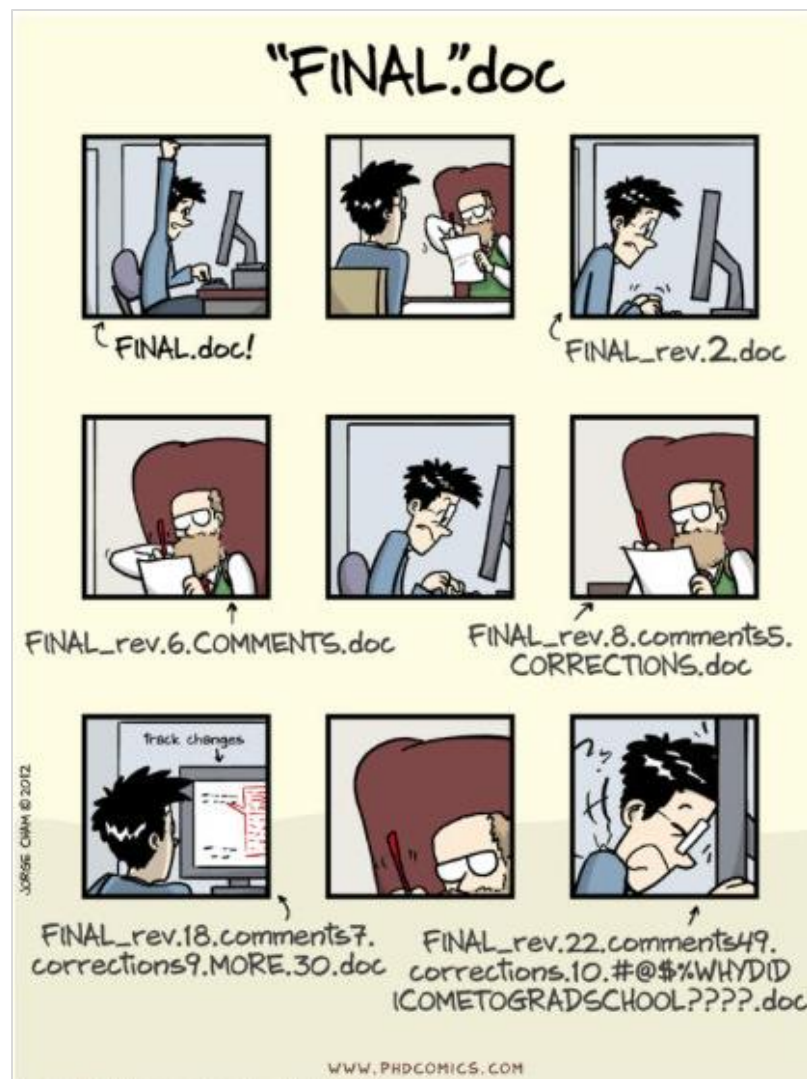
# Git – система контроля версий (VCS)

*Есть другие программы с интерфейсом  
Но внутри тот же Git  
(GitKraken , GitHub Desktop)*



# Git – система контроля версий (VCS)

Есть другие программы с интерфейсом  
Но внутри тот же Git  
(GitKraken, GitHub Desktop)



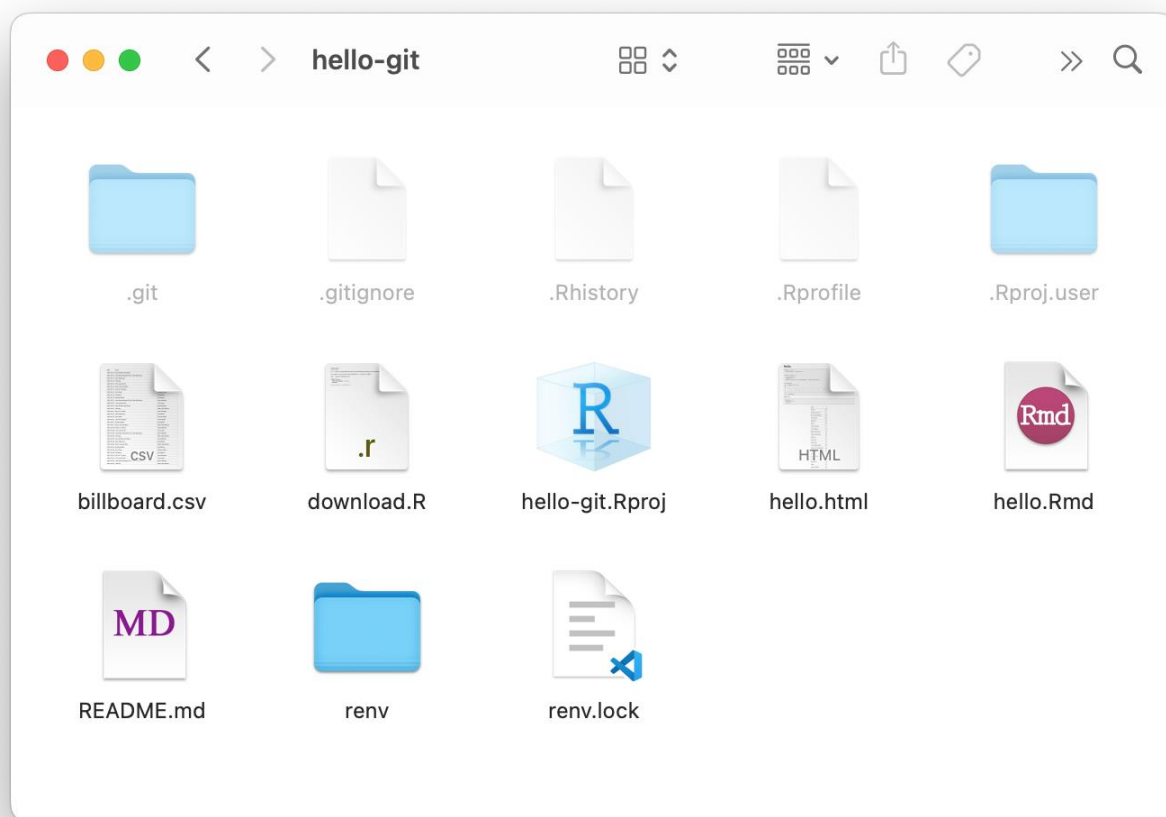
# Зачем

- Всегда есть резервная копия
- Легко вернуться к исходному состоянию
- Итеративный процесс работы
- Работа в команде (ревью)



# Git – система контроля версий (VCS)

*Есть другие программы с интерфейсом  
Но внутри тот же Git  
(GitKraken, GitHub Desktop)*



← Репозиторий (локальный)



# GitHub – хостинг репозиториев

Есть другие хостинги  
GitLab, Gitea, Bitbucket, внутри организации

yoursdearboy

Added bingo cards to git ignore

78d0032 · 4 days ago · 20 Commits

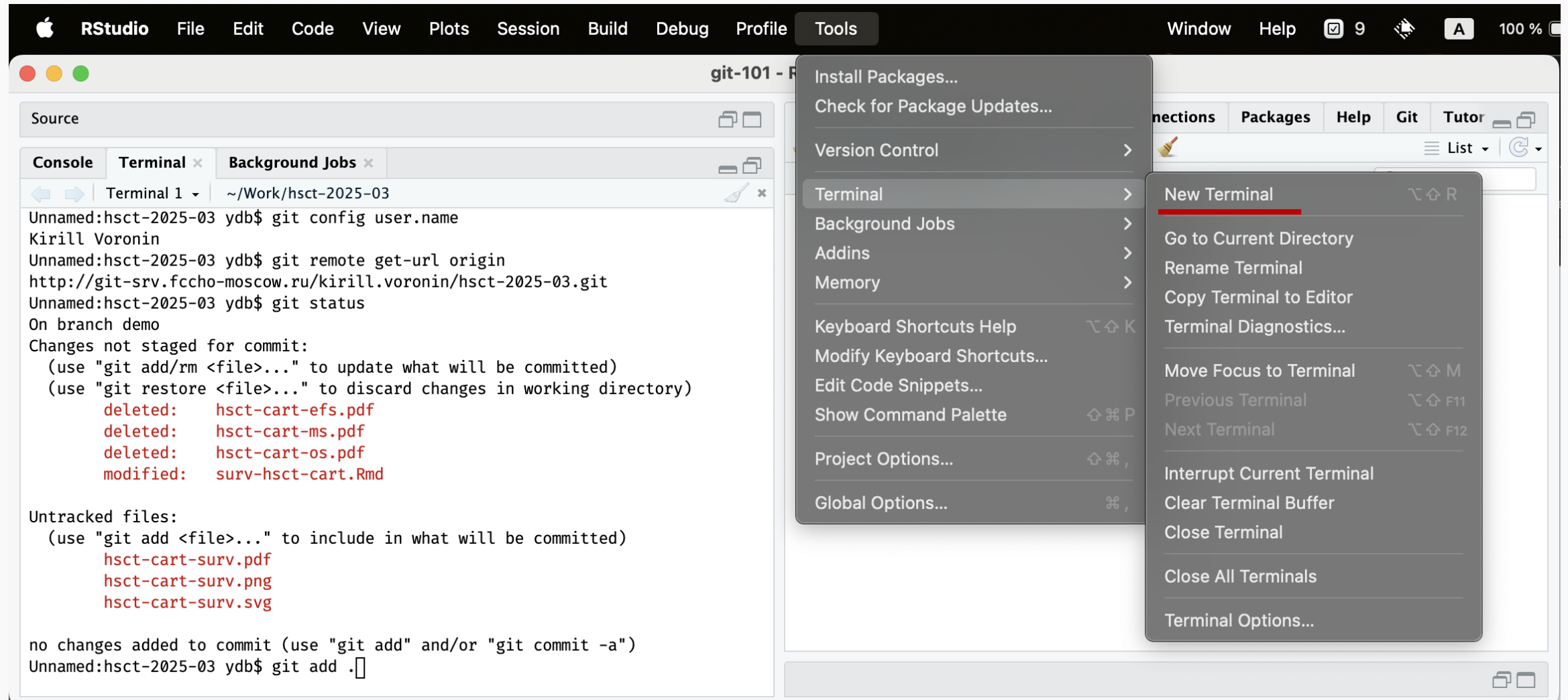
|                 |                                                        |            |
|-----------------|--------------------------------------------------------|------------|
| renv            | Created R project with renv                            | last week  |
| .Rprofile       | Created R project with renv                            | last week  |
| .gitignore      | Added bingo cards to git ignore                        | 4 days ago |
| README.md       | Create README.md                                       | 4 days ago |
| artists.txt     | Ooops Cyrillic not supported                           | 4 days ago |
| billboard.csv   | Downloaded and transformed billboard top 100 to top 10 |            |
| bingo.R         | Added script to play bingo (#4)                        |            |
| data.R          | Downloaded and transformed billboard top 100 to top 10 |            |
| hello-git.Rproj | Created R project with renv                            |            |
| hello.Rmd       | Sampled top artists                                    |            |
| hello.html      | Sampled top artists                                    |            |
| renv.lock       | Added script to play bingo (#4)                        |            |



Репозиторий (remote)

|                                                                                                                          |                                   |         |  |    |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------|---------|--|----|
| Sampled top artists                                                                                                      | yoursdearboy committed 3 days ago | 8529589 |  | <> |
| Created packages snapshot                                                                                                | yoursdearboy committed 3 days ago | fa98548 |  | <> |
| Merge branch 'main' of <a href="https://github.com/yoursdearboy/hello-git">https://github.com/yoursdearboy/hello-git</a> | yoursdearboy committed 3 days ago | d32b7c3 |  | <> |
| Downloaded and transformed billboard top 100 to top 10                                                                   | yoursdearboy committed 3 days ago | bcd3db  |  | <> |
| Create README.md                                                                                                         | yoursdearboy authored 3 days ago  | 3990f56 |  | <> |
| Commits on Dec 3, 2024                                                                                                   |                                   |         |  |    |
| Draw some sample                                                                                                         | yoursdearboy committed 5 days ago | 6e92f7d |  | <> |
| Created R project with renv                                                                                              | yoursdearboy committed 5 days ago | ec229cb |  | <> |

# RStudio



# Терминал

```
$ command subcommand --long-argument -s arg1 arg2 ...
```

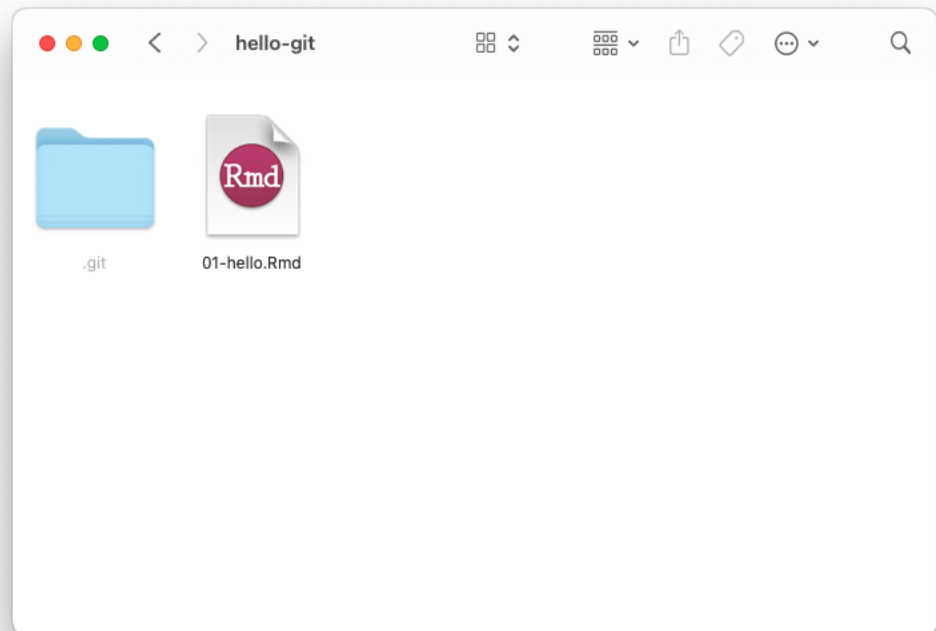
```
$ git config --global user.name "Kirill Voronin"
```

```
$ git config --global user.email imvoroninkir@gmail.com
```

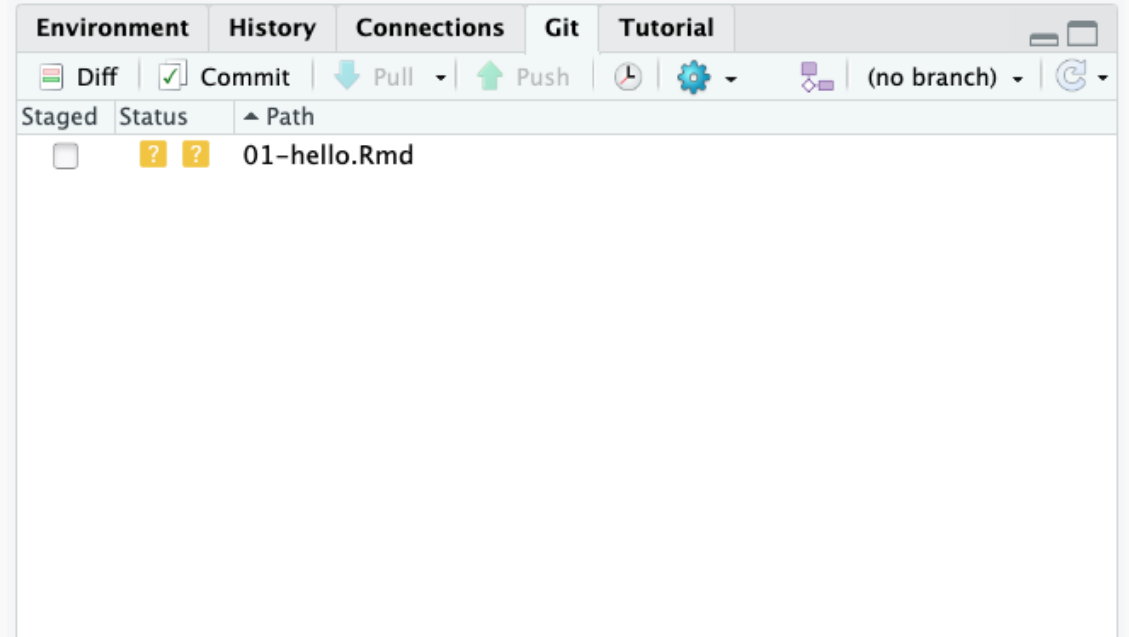
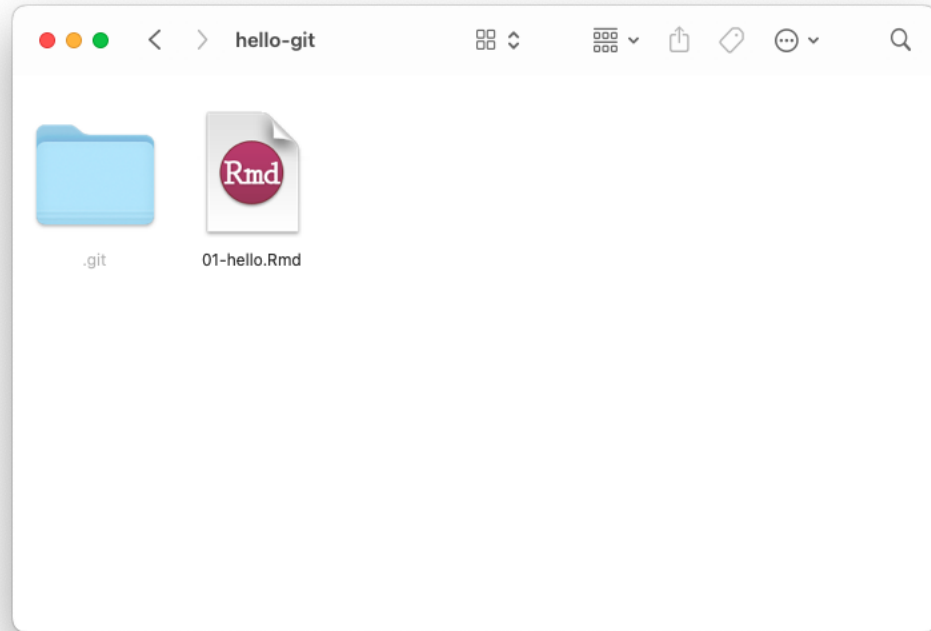
```
$ git remote add origin https://github.com/user/repo.git
```

```
$ git push -u origin main
```

# Рабочая область (Working tree)

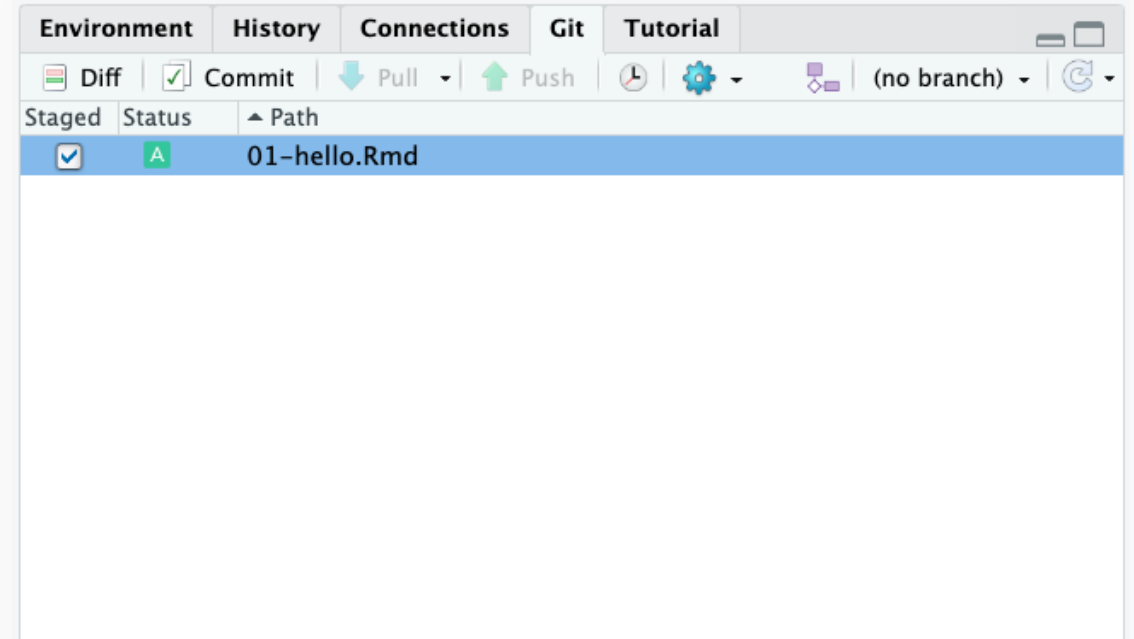
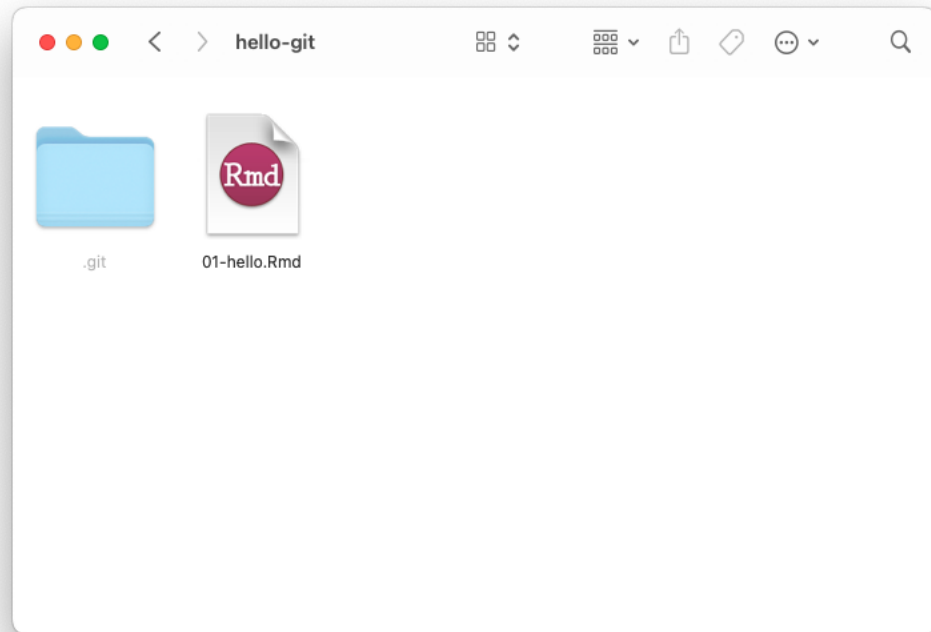


# Рабочая область (Working tree)



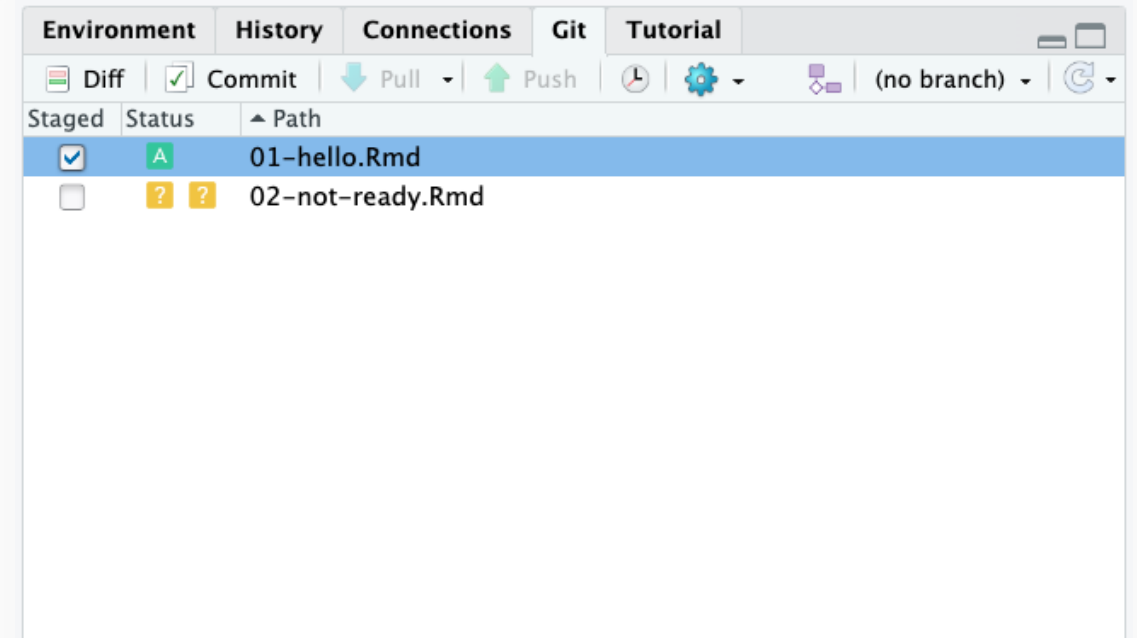
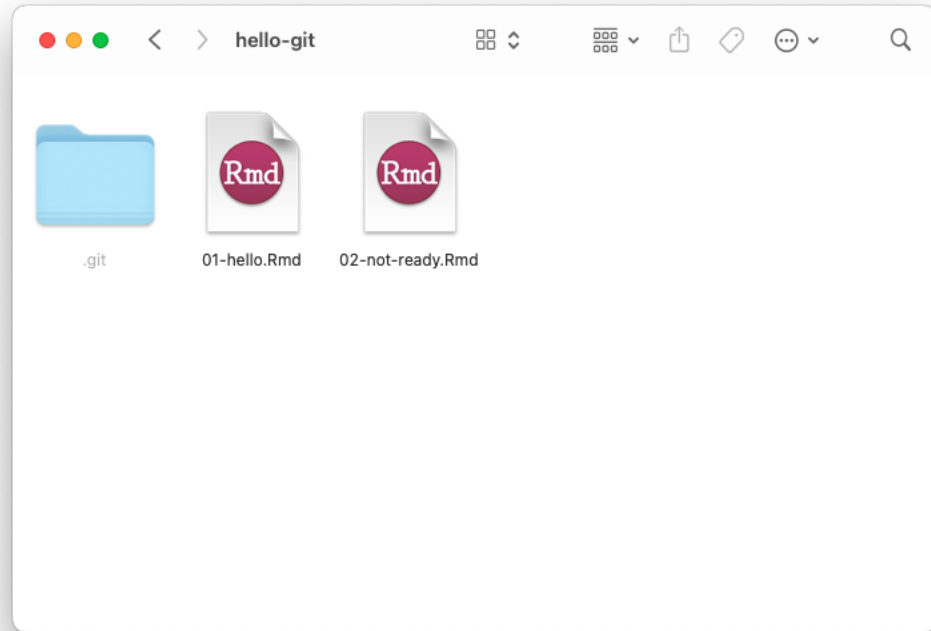
```
$ git status
```

# Рабочая область → Staging



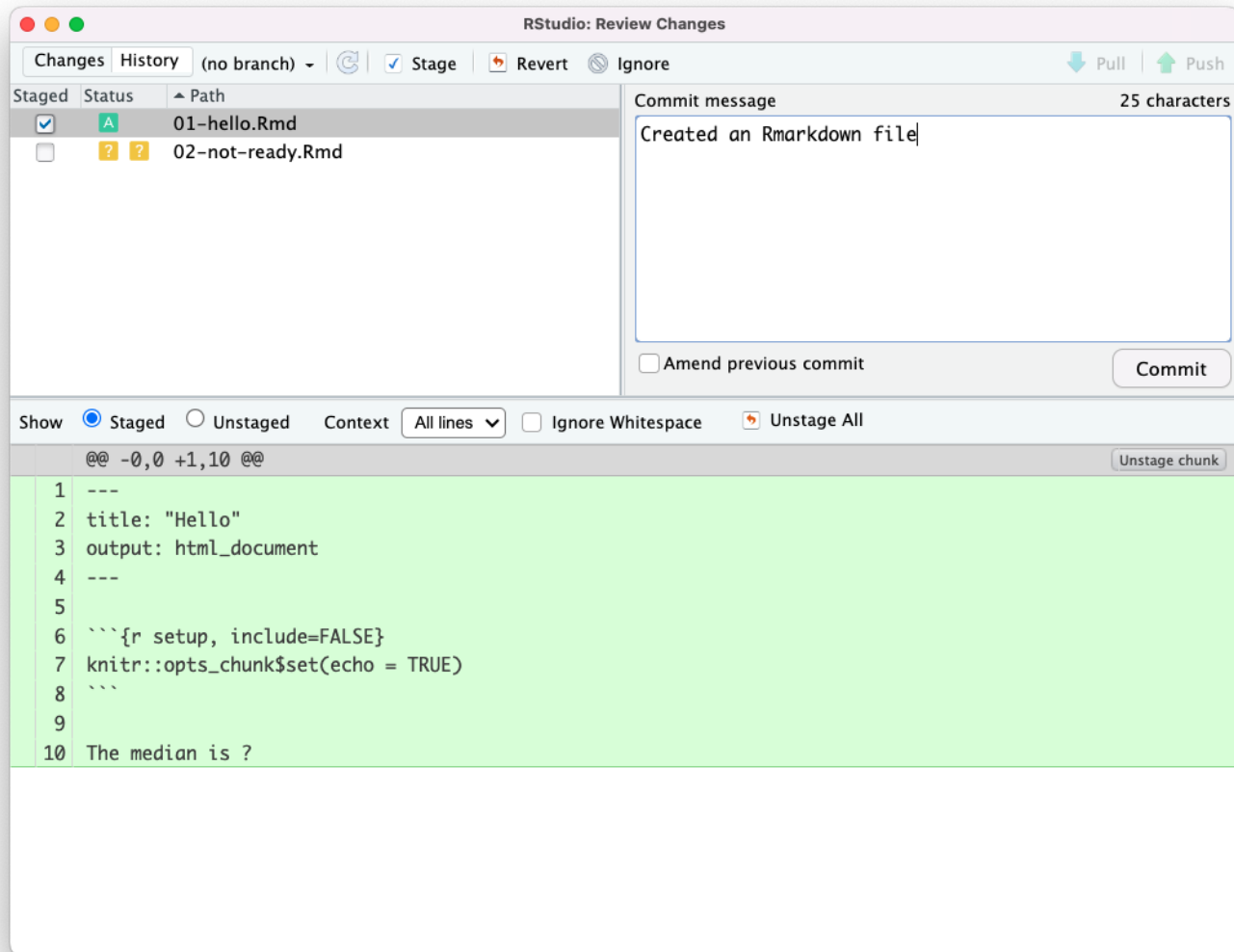
```
$ git add 01-hello.Rmd
```

# Рабочая область ➡ Staging



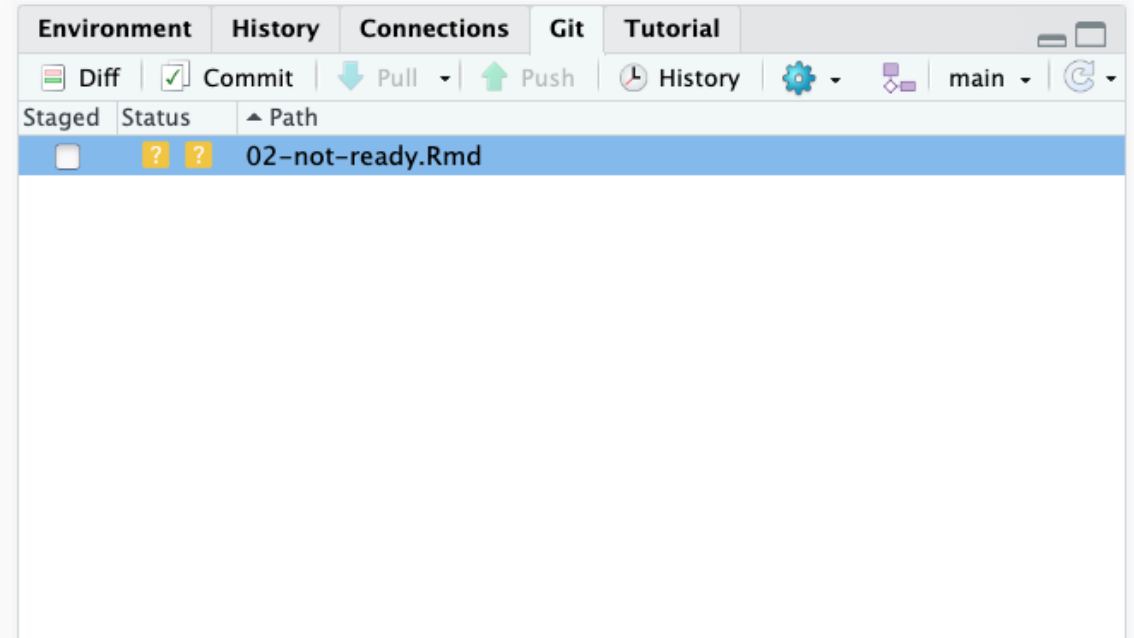
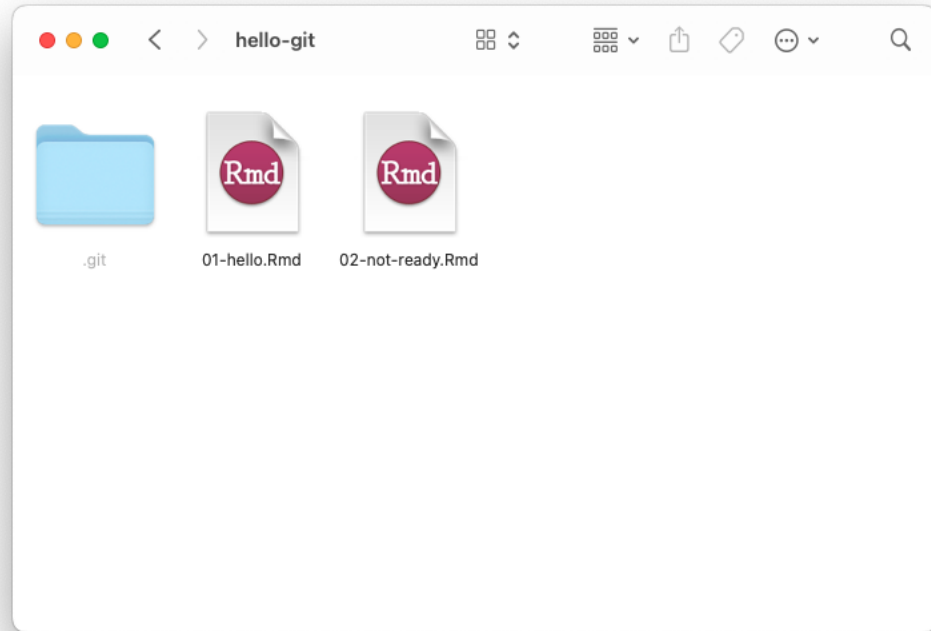
```
$ git status
```

# Фиксация изменений (Commit)



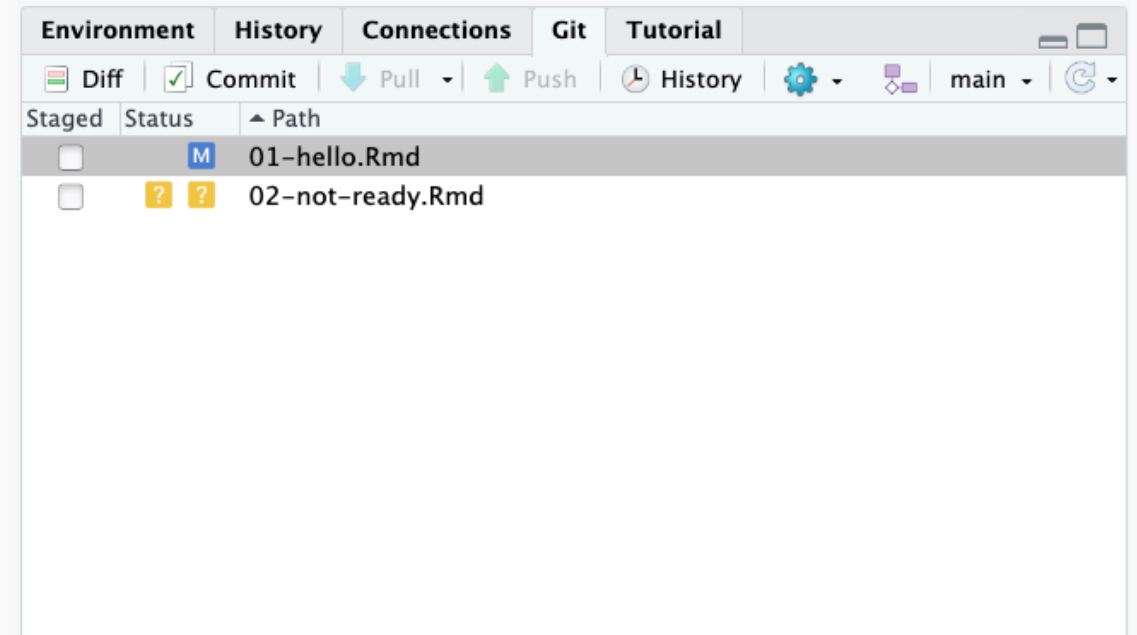
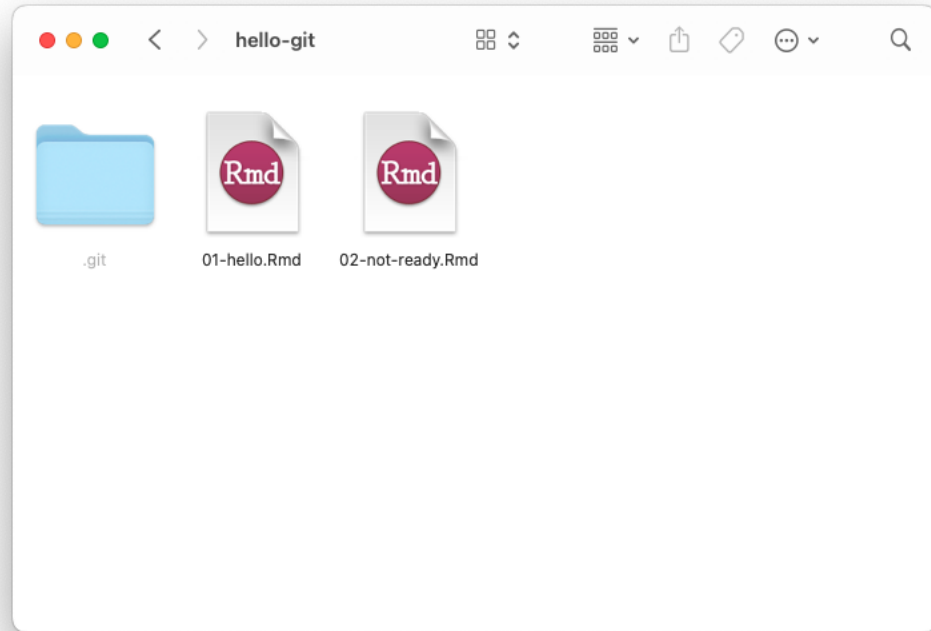
```
$ git commit -m "Created an Rmarkdown file"
```

# Рабочая область (Working tree)



```
$ git status
```

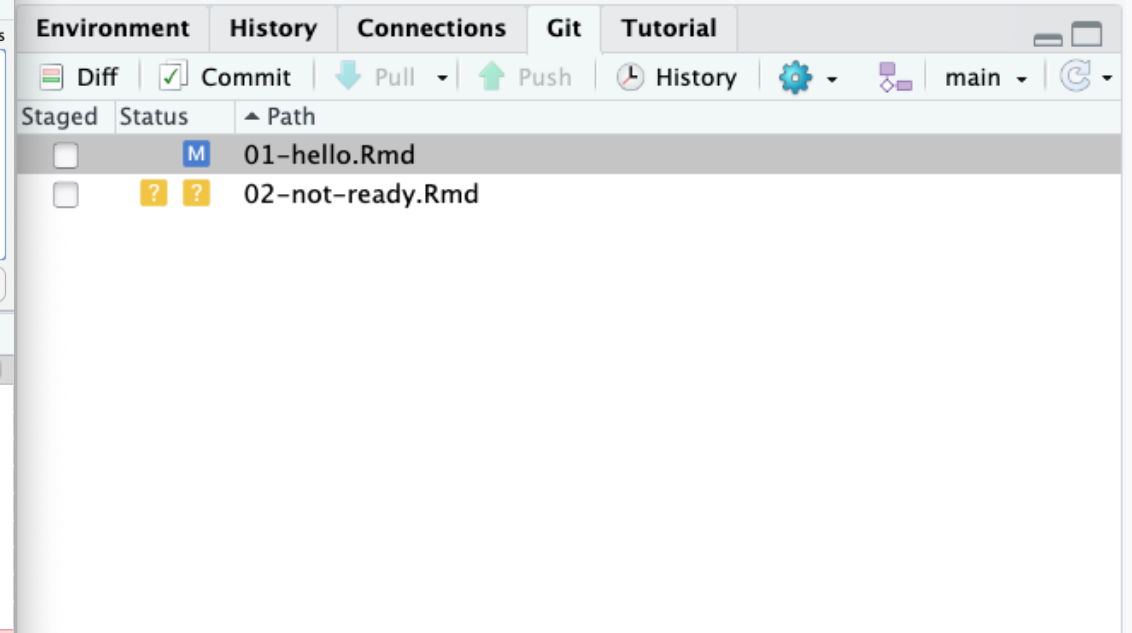
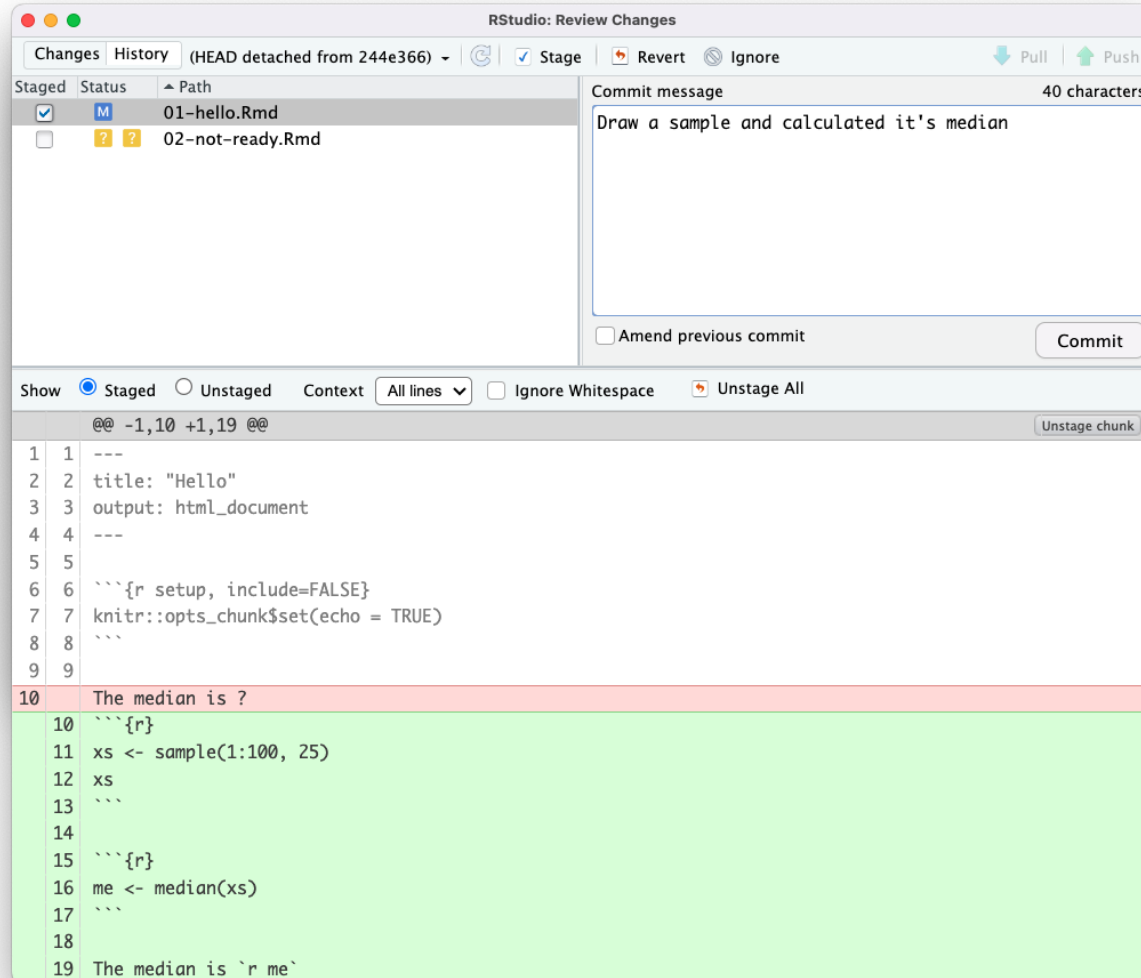
# Рабочая область (Working tree)



```
$ git add 01-hello.Rmd
```

```
$ git status
```

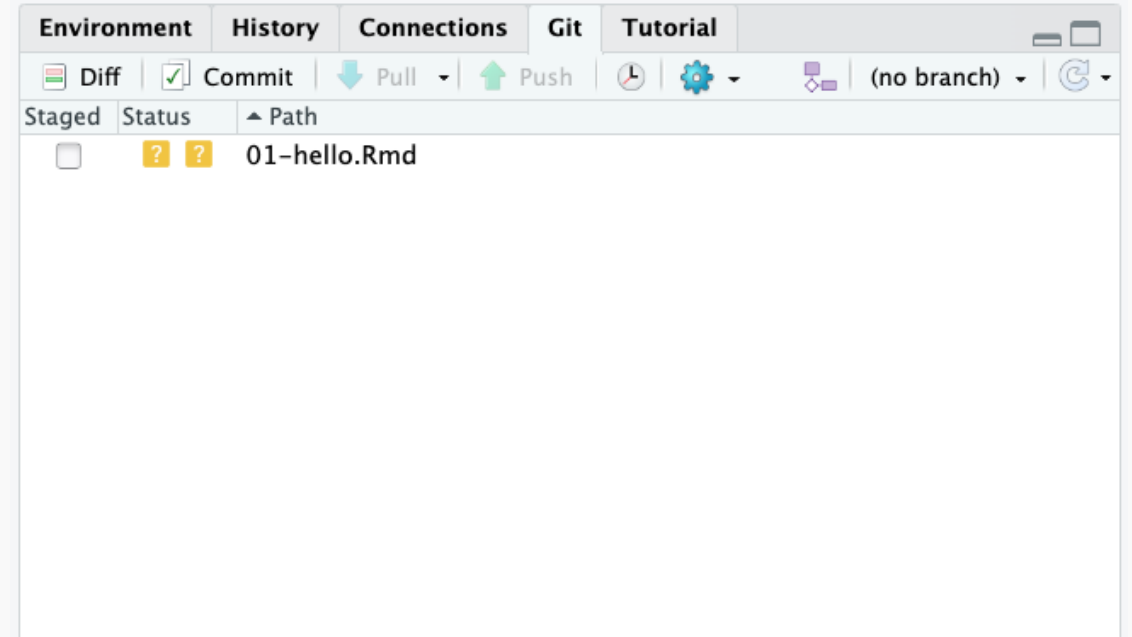
# Фиксация изменений (Commit)



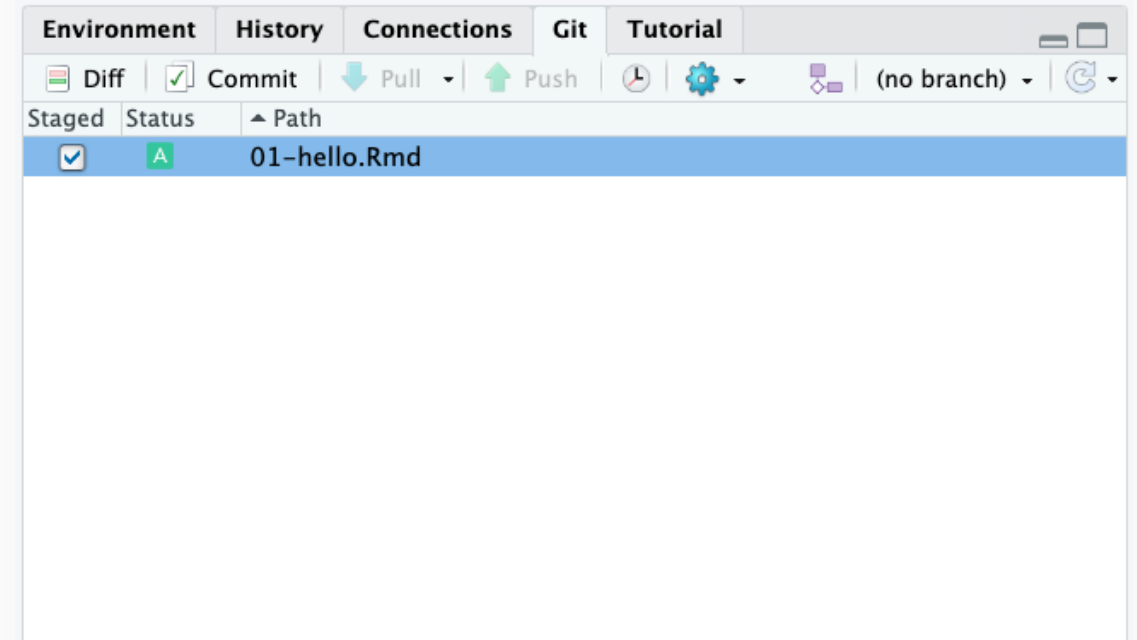
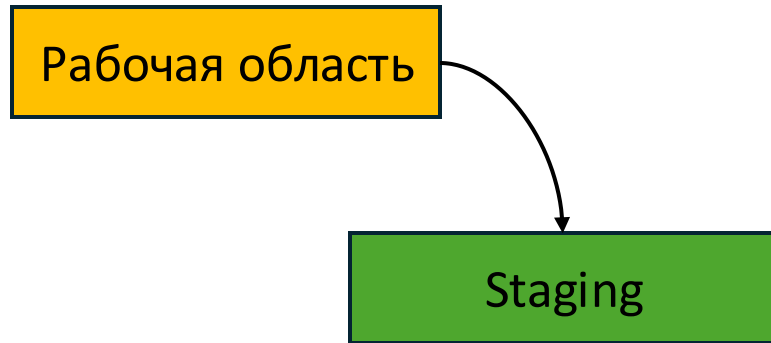
```
$ git commit -m "Draw a sample and calculated it's median"
```

# Итеративный процесс работы

Рабочая область

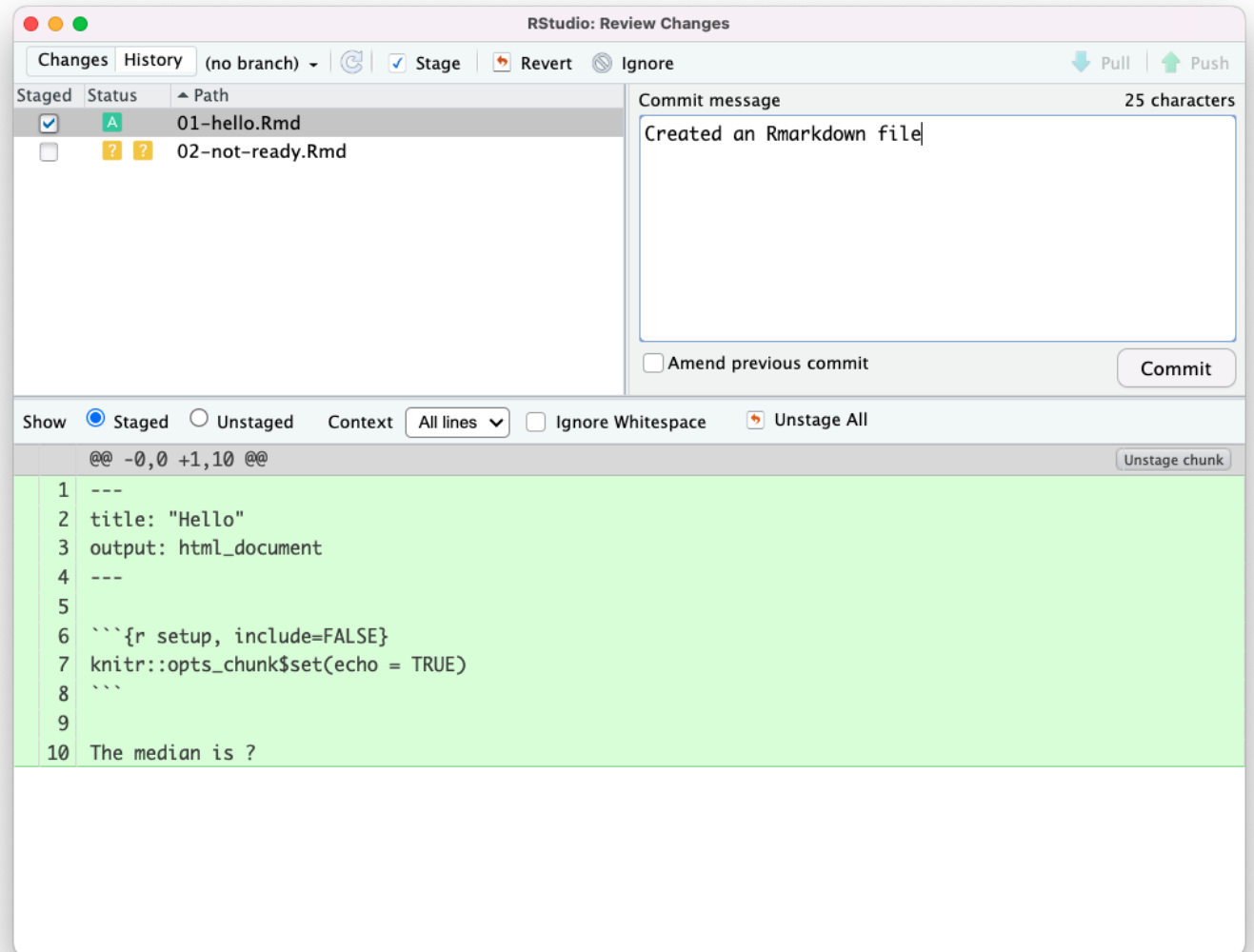
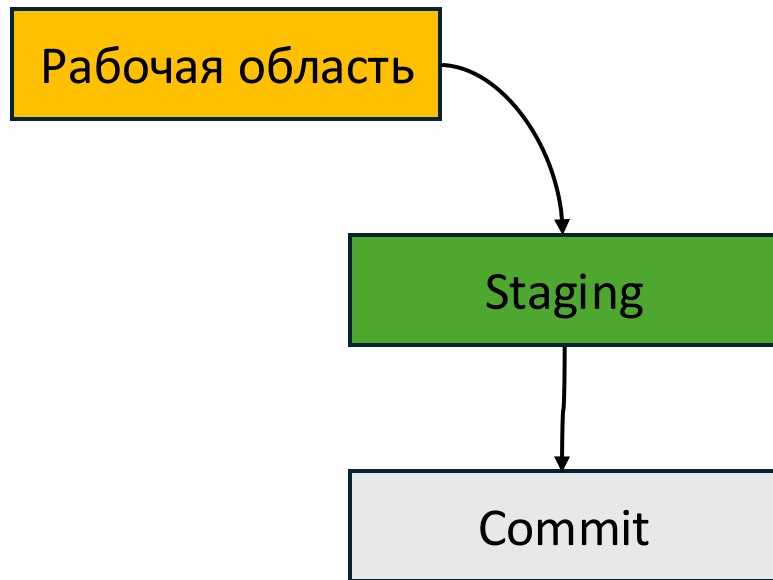


# Итеративный процесс работы



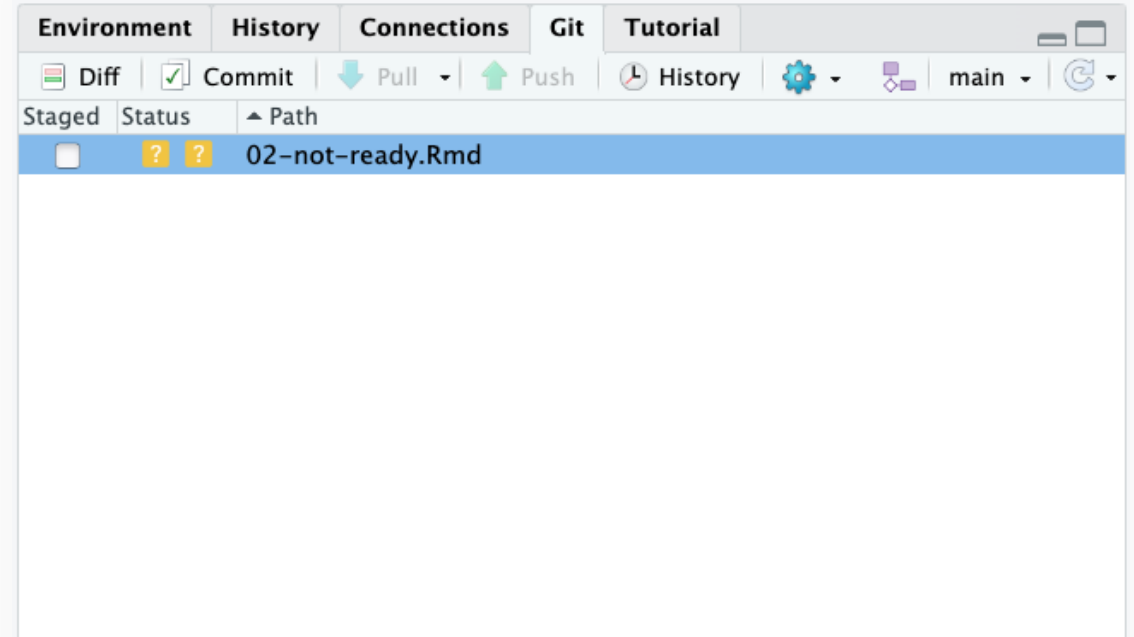
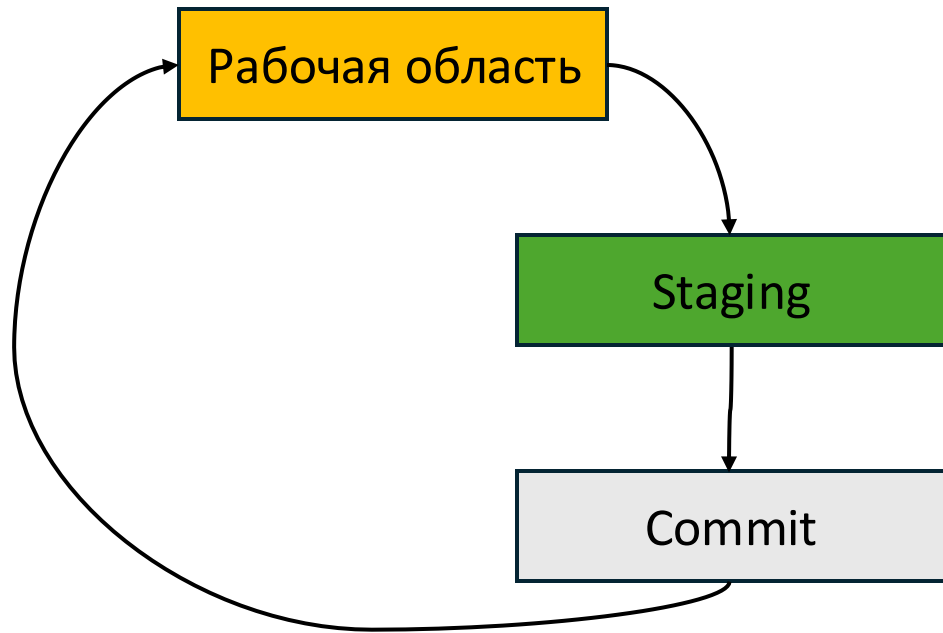
```
$ git add 01-hello.Rmd
```

# Итеративный процесс работы

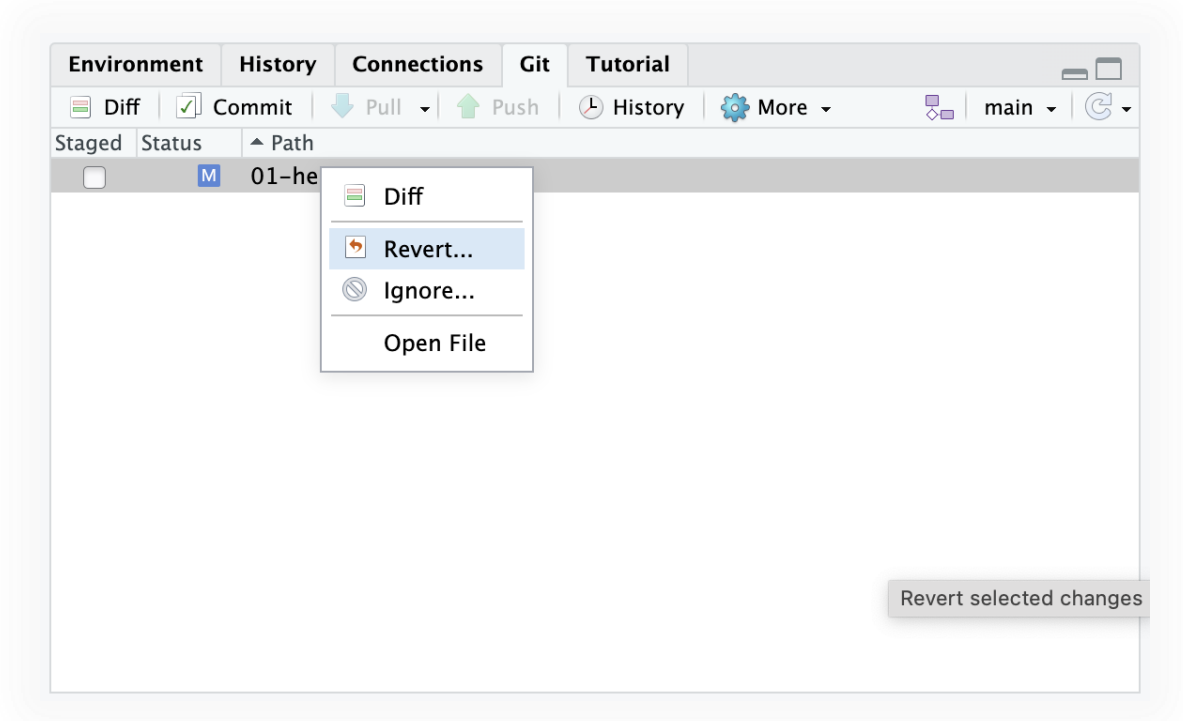
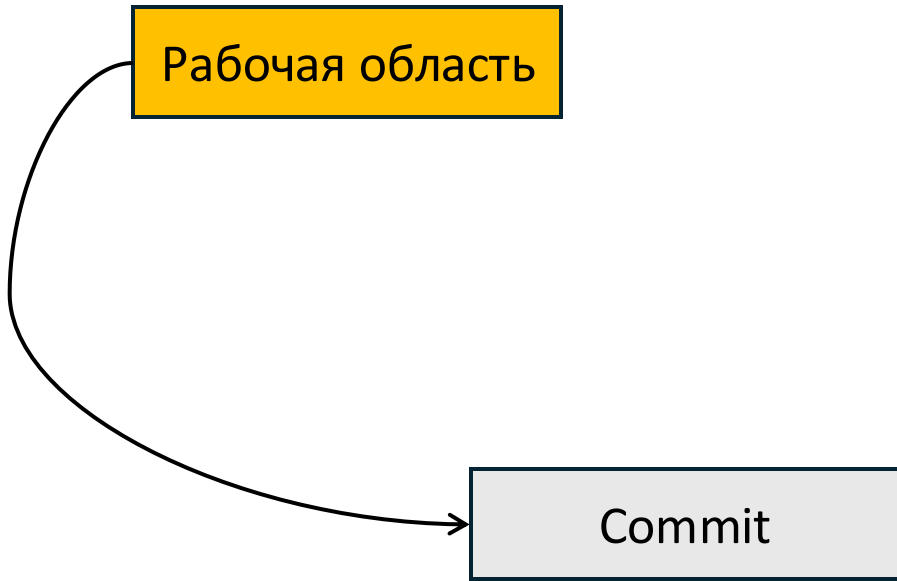


```
$ git commit -m "Created an Rmarkdown file"
```

# Итеративный процесс работы



# Итеративный процесс работы



```
$ git diff 01-hello.Rmd  
$ git restore 01-hello.Rmd
```

# Журнал изменений (log)

The screenshot shows the RStudio 'Review Changes' interface. At the top, there's a navigation bar with 'Changes' and 'History' tabs, and a search bar. Below this is a table of commits. The selected commit is 'HEAD -> refs/heads/main: Sampled top artists' by Kirill Voronin, dated 2024-12-11, with SHA 8ebaba3e. Below the table, there's a section for the selected commit, showing its SHA, author, date, subject, and parent. At the bottom, there's a diff view for the file '01-hello.Rmd', showing line-by-line changes with green highlights for additions.

| Subject                                                       | Author                         | Date (UTC) | SHA      |
|---------------------------------------------------------------|--------------------------------|------------|----------|
| HEAD -> refs/heads/main: Sampled top artists                  | Kirill Voronin <yoursdearboy@> | 2024-12-11 | 8ebaba3e |
| Created packages snapshot                                     | Kirill Voronin <yoursdearboy@> | 2024-12-11 | 943a9704 |
| Merge branch 'main' of https://github.com/yoursdearboy/hello- | Kirill Voronin <yoursdearboy@> | 2024-12-11 | 7af7b404 |
| Downloaded and transformed billboard top 100 to top 10        | Kirill Voronin <yoursdearboy@> | 2024-12-11 | 11ecca28 |
| readme: Create README.md                                      | Kirill Voronin <yoursdearboy@> | 2024-12-11 | 3153ad90 |
| Draw a sample and calculated it's median                      | Kirill Voronin <yoursdearboy@> | 2024-12-11 | 343b6f67 |
| Created an Rmarkdown file                                     | Kirill Voronin <yoursdearboy@> | 2024-12-11 | bcce4337 |

SHA: 8ebaba3e13780325feab9d6f930a2b74ce56426a  
Author: Kirill Voronin <yoursdearboy@gmail.com>  
Date (UTC): 2024-12-11 14:41  
Subject: Sampled top artists  
Parent: 943a9704c8c60983fa04fb75f16a8d5428e6d2af

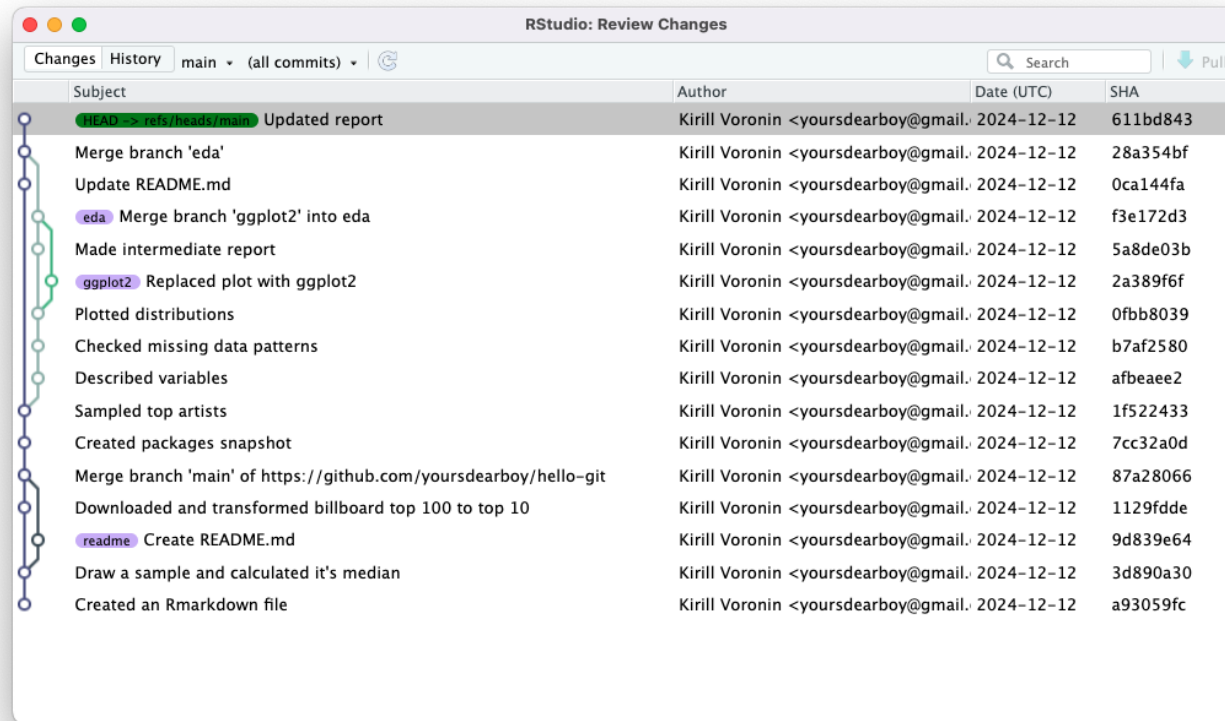
01-hello.Rmd

@@ -4,9 +4,19 @@ output: html\_document

```
4 4 ---
5 5
6 6 ```{r setup, include=FALSE}
7 7 library(dplyr)
8 8 library(readr)
9 9
7 10 knitr::opts_chunk$set(echo = TRUE)
8 11 ```
9 12
13 13 ```{r data, include=FALSE}
14 14 top <- read_csv("billboard.csv") |>
15 15   select(date = "Date",
16 16          song = "Song",
17 17          artist = "Artist")
```

```
$ git log
```

# Ветви (branch)



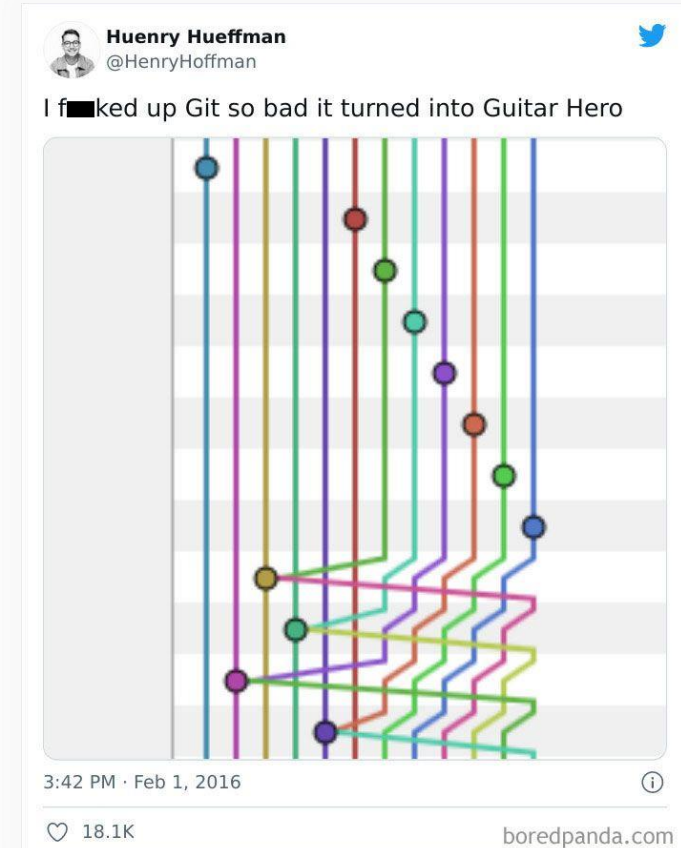
```
$ git branch
```

# Ветви (branch)

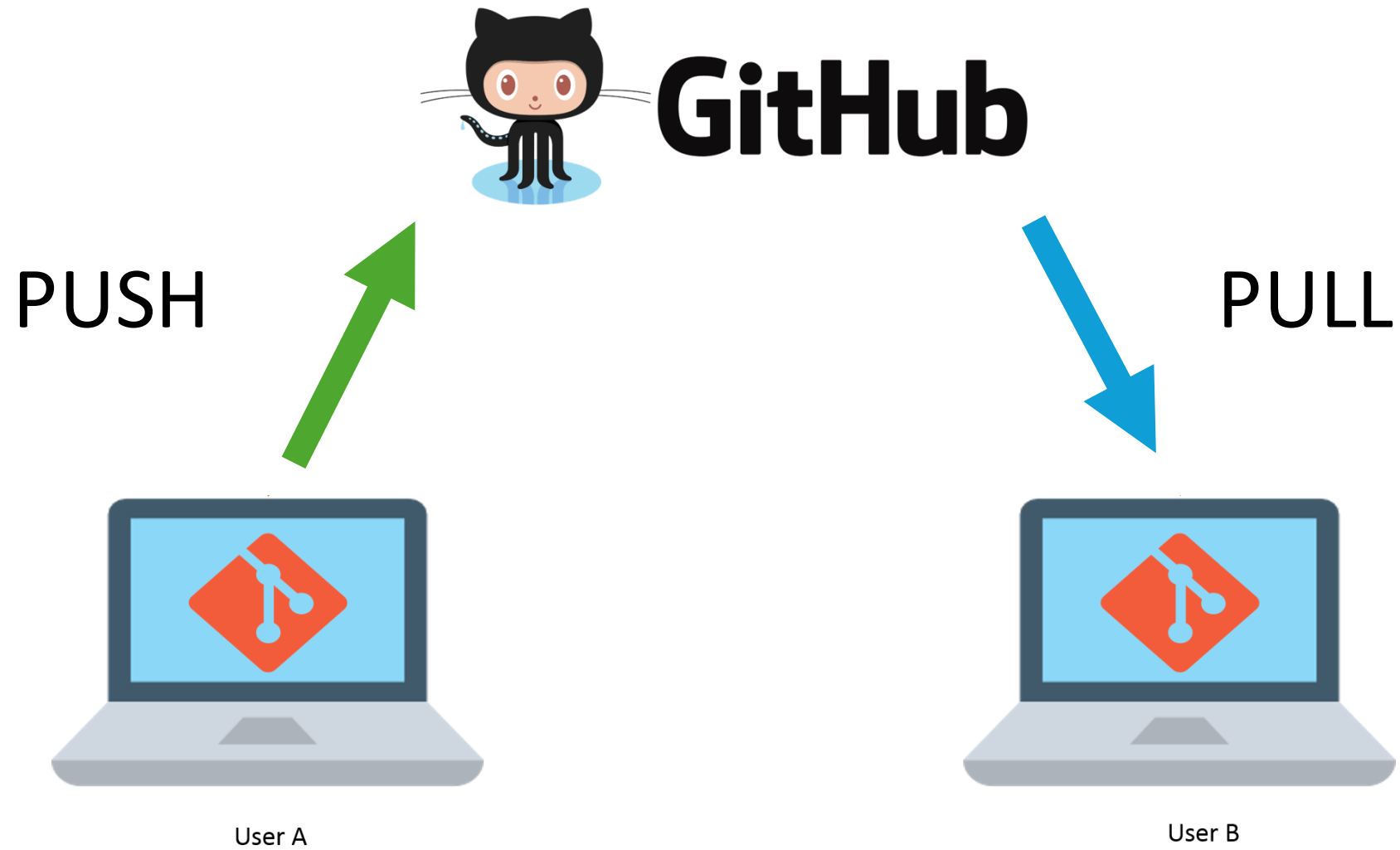
RStudio: Review Changes

Changes History main (all commits) Search Pull

| Subject                                                          | Author                              | Date (UTC) | SHA      |
|------------------------------------------------------------------|-------------------------------------|------------|----------|
| HEAD -> refs/heads/main Updated report                           | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | 611bd843 |
| Merge branch 'eda'                                               | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | 28a354bf |
| Update README.md                                                 | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | 0ca144fa |
| eda Merge branch 'ggplot2' into eda                              | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | f3e172d3 |
| Made intermediate report                                         | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | 5a8de03b |
| ggplot2 Replaced plot with ggplot2                               | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | 2a389f6f |
| Plotted distributions                                            | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | 0fbb8039 |
| Checked missing data patterns                                    | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | b7af2580 |
| Described variables                                              | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | afbeaee2 |
| Sampled top artists                                              | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | 1f522433 |
| Created packages snapshot                                        | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | 7cc32a0d |
| Merge branch 'main' of https://github.com/yoursdearboy/hello-git | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | 87a28066 |
| Downloaded and transformed billboard top 100 to top 10           | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | 1129fdde |
| readme Create README.md                                          | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | 9d839e64 |
| Draw a sample and calculated it's median                         | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | 3d890a30 |
| Created an Rmarkdown file                                        | Kirill Voronin <yoursdearboy@gmail> | 2024-12-12 | a93059fc |



\$ git branch



```
$ git push  
$ git pull
```

# Pull Request (PR)

pharmaverse / ggsurvfit

<> Code

Issues 20

**Pull requests 4**

Discussions

Actions

Projects

Security

Insights

Type / to search

fixed KMunicate legend position in gallery.Rmd #223

<> Code

Open

DanChaltiel wants to merge 3 commits into `pharmaverse:main` from `DanChaltiel:patch-2`

Conversation 1

Commits 3

Checks 11

Files changed 2

+28 -10

 **DanChaltiel** commented on Sep 19

Contributor

Hi,  
  
As you can see in <https://www.danielsjoberg.com/ggsurvfit/articles/gallery.html#kmunicate>, the legend is not inside the plot as it should be.  
  
This is because the theme lacks `legend.position="inside"`.  
  
This very minor PR should fix it.

Reviewer Checklist (if item does not apply, mark as complete)

☒ Ensure all package dependencies are installed by running `renv::install()`

☒ PR branch has pulled the most recent updates from master branch. Ensure the pull request branch and your local version match and both have the latest updates from the master branch.

☒ If a new function was added, function included in `_pkgdown.yml`

☒ If a bug was fixed, a unit test was added for the bug check

☒ Run `pkgdown::build_site()`. Check the R console for errors, and review the rendered website.

☒ Overall code coverage remains >99.5%. Review coverage with `withr::with_envvar(list(CI = TRUE), code = devtools::test_coverage())`. Begin in a fresh R session without any packages loaded.

☒ R CMD Check runs without errors, warnings, and notes

☒ `usethis::use_spell_check()` runs with no spelling errors in documentation

When the branch is ready to be merged into master:

☐ Update `NEWS.md` with the changes from this pull request under the heading "`# ggsurvfit (development version)`". If there is an issue associated with the pull request, reference it in parentheses at the end update (see `NEWS.md` for examples).

☐ Increment the version number using `usethis::use_version(which = "dev")`

☐ Run `usethis::use_spell_check()` again

☐ Approve Pull Request

Reviewers

No reviews

Still in progress? Learn about draft PRs

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications

Customize

Subscribe

You're not receiving notifications from this thread.

2 participants



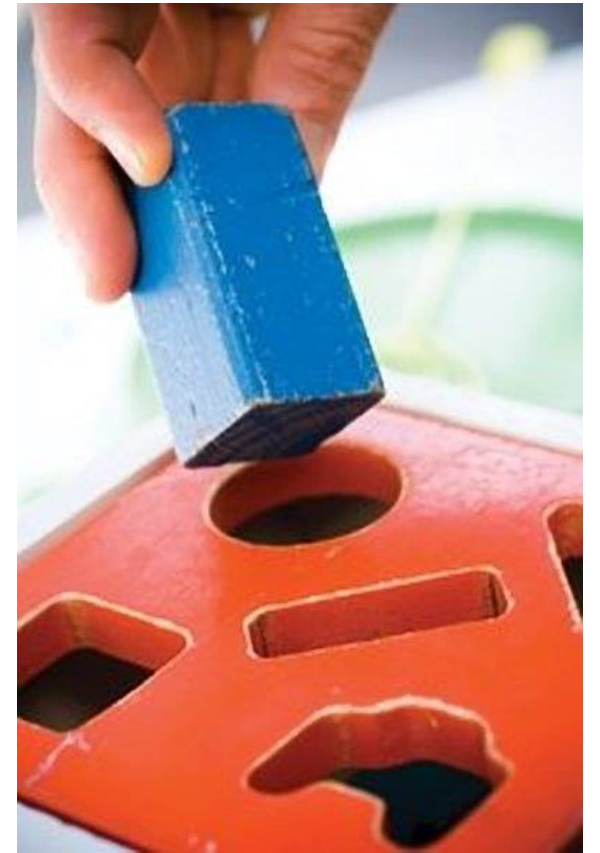
# Не копите изменения в рабочей области

| Environment                         | History | Connections     | Git | Tutorial |
|-------------------------------------|---------|-----------------|-----|----------|
| Diff                                | Commit  |                 |     |          |
| Staged                              | Status  | Path            |     |          |
| <input type="checkbox"/>            | ? ?     | FINAL.doc       |     |          |
| <input checked="" type="checkbox"/> | A       | extras.R        |     |          |
| <input checked="" type="checkbox"/> | A       | hello-copy1.Rmd |     |          |
| <input checked="" type="checkbox"/> | A       | hello-copy1.pdf |     |          |
| <input checked="" type="checkbox"/> | M       | hello.Rmd       |     |          |
| <input checked="" type="checkbox"/> | M       | hello.html      |     |          |
| <input checked="" type="checkbox"/> | A       | image01.png     |     |          |
| <input type="checkbox"/>            | ? ?     | image02.png     |     |          |
| <input type="checkbox"/>            | ? ?     | image03.png     |     |          |
| <input type="checkbox"/>            | ? ?     | test1.R         |     |          |

# git push --force

Вы пытаетесь переписать чужие изменения.

Вы пытаетесь переписать свои изменения,  
но если их уже кто-то запулил –  
все сгорит 💀



# Все сломалось

- Закройте RStudio и переименуйте папку
- Склонируйте репозиторий заново
- Внесите изменения из старой папки
- Удалите старую папку



# Структура репозитория

- data/surgery.csv
- data/chemo.csv
- data.R
- fig/02-os.pdf
- fig/03-os-cox-forestplot.pdf
- misc.R
- 01-EDA.Rmd (01-EDA.html)
- 02-plot.Rmd (02-plot.html)
- 03-model.Rmd (03-model.html)
- renv.lock
- README.md

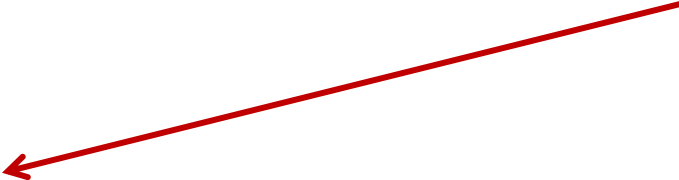
Данные тоже версионированы.  
Есть история изменений.

Результат тоже.

# Структура репозитория

- data/surgery.csv
- data/chemo.csv
- data.R
- fig/02-os.pdf
- fig/03-os-cox-forestplot.pdf
- misc.R
- stat\_stepribbon.R
- 01-EDA.Rmd (01-EDA.html)
- 02-plot.Rmd (02-plot.html)
- 03-model.Rmd (03-model.html)
- renv.lock
- README.md

Данные тоже версионироваться.  
Есть история изменений.



Результат тоже.



```
source("data.R")  
source("misc.R")
```

# README.md

## # Название проекта

Краткое описание

## ## Данные

В файле `[imdb250.csv](imdb250.csv)` содержится  
250 самых оцениваемых фильмов с 1980 по 2024 год.

## ## Анализ выживаемости

Краткое описание и ссылка на файл с анализом. Общая выживаемость проанализирована при помощи `[регрессии Кокса, построены скорректированные (adjusted) кривые выживаемости](03-mod.Rmd)` при помощи `survexp[^1]`.

## ## Ссылки

<sup>[^1]</sup>: Adjusted Survival Curves. Therneau, Crowson, Atkinson, 2015

# Ссылки

- [Good enough practices in scientific computing, Data Carpentry](#)
- [Best Practices for Scientific Computing, Data Carpentry](#)
- [Happy Git and GitHub for the useR](#)
- [What They Forgot to Teach You About R](#)